TEAM HOFFNANG — ALGORITHM REFERENCE NOTEBOOK Soyebuzaman Naim | Team Hoffnang (Southeast University) | C++17

1. CONTEST SETUP & TEMPLATE

Fast I/O CP Template `cpp`

Disables sync, unties cin, defines 64-bit int, standard loop shorthands.

```
#include <bits/stdc++.h>
using namespace std;
#define int long long
#define endl '\n'
#define all(x) (x).begin(), (x).end()
#define rall(x) (x).rbegin(), (x).rend()
#define sz(x) (int)(x).size()
void fast_io() { ios::sync_with_stdio(false); cin.tie(nullptr); }
signed main() {
  fast_io();
  int t; cin >> t;
  while (t--) {
    int n; cin >> n;
    vector<int> v(n);
    for (int &x : v) cin >> x;
  }
}
```

Custom Sort & Comparators `custom_sort` | `O(N log N)`

Lambda, function, struct operator<, index/argsort, and priority_queue comparator.

```
// 1. Lambda Comparator (Sort in-place without external function):
sort(v.begin(), v.end(), [](const auto& a, const auto& b) {
  if (a.first != b.first) return a.first < b.first; // Ascending by first
  return a.second > b.second;                       // Descending by second
});

// 2. Free Function Comparator:
bool comp(const pair<int, int>& a, const pair<int, int>& b) {
  return a.first < b.first;
}
sort(v.begin(), v.end(), comp);

// 3. Struct with Internal operator<:
struct Item {
  int val, id;
  bool operator<(const Item& o) const {
    if (val != o.val) return val < o.val;
    return id < o.id;
  }
};

// 4. Index Sorting (argsort - sort original indices by values):
vector<int> p(n); iota(p.begin(), p.end(), 0);
sort(p.begin(), p.end(), [&](int i, int j) { return a[i] < a[j]; });

// 5. Comparator Struct for Priority Queue (Min-Heap by first):
struct ComparePQ {
  bool operator()(const pair<int, int>& a, const pair<int, int>& b) {
    return a.first > b.first; // Greater-than creates Min-Heap!
  }
};
priority_queue<pair<int, int>, vector<pair<int, int>>, ComparePQ> pq;
```

GCC Optimization Pragmas `pragma`

Hardware vectorization and loop unrolling for tight time limits (1.0s).

```
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

__int128 Fast I/O & Output `int128` | ~3.4×10^{38}

Native 128-bit integer with custom stream parser and recursive print.

```
__int128 read128() {
  __int128 x = 0; int f = 1; char ch = getchar();
  while (ch < '0' || ch > '9') { if (ch == '-') f = -1; ch = getchar(); }
  while (ch >= '0' && ch <= '9') { x = x * 10 + ch - '0'; ch = getchar(); }
  return x * f;
}
void print128(__int128 x) {
  if (x < 0) { putchar('-'); x = -x; }
  if (x > 9) print128(x / 10);
  putchar(x % 10 + '0');
}
```

2. C++ BASICS & STL ESSENTIALS

Types, Overflow & Float Comparisons

Type	Bytes	Range	Rule
`int`	4	$[-2\times10^9, 2\times10^9]$	Default loop variables
`long long`	8	$[-9\times10^{18}, 9\times10^{18}]$	Sums, products: 1LL*a*b
`double`	8	15-17 digits	`fabs(a-b) < 1e-9`

```
int a = 1e6, b = 1e6;
long long ans = 1LL * a * b; // 1LL prevents 32-bit overflow!
cout << fixed << setprecision(9) << ans << "\n";
```

C++ <cmath> Functions (exp, ln, root, sin, cos)

`cmath`

e^x, natural log ln(x), square/cube/k-th roots, trigonometry, and angle conversions.

```
const double PI = acos(-1.0); // Exact π: 3.141592653589
793

// 1. Exponential (e^x) & Powers:
double e_to_x   = exp(x);              // e^x (for long
double use expl(x))
double power    = pow(base, exp);      // base^exp (e.g.
pow(2.0, 10.0) = 1024.0)

// 2. Natural Logarithm (ln) & Logarithms:
double natural  = log(x);              // ln(x) (natural
log, base e)
double log_10   = log10(x);            // log10(x) (base
10)
double log_2    = log2(x);             // log2(x) (base
2)
double log_b    = log(x) / log(b);     // log_b(x) in ar
bitrary base b

// 3. Roots (Square, Cube & k-th Root):
double sq_root  = sqrt(x);             // √x (Square roo
t: x >= 0)
double cb_root  = cbrt(x);             // ∛x (Cube root:
negative inputs allowed!)
double kth_root = pow(x, 1.0 / k);     // k-th root of
x: x^(1/k)

// 4. Trigonometry: sin, cos, tan (Input MUST be in RADI
ANS!):
double deg = 60.0;
double rad = deg * (PI / 180.0);       // Degrees to Rad
ians!
double s   = sin(rad);                 // sin(60°) = √3
/ 2 ≈ 0.866025
double c   = cos(rad);                 // cos(60°) = 0.5
double t   = tan(rad);                 // tan(60°) ≈ 1.7
3205

// 5. Inverse Trigonometry & Angles:
double angle_rad = atan2(y, x);        // Angle in radia
ns [-π, π] from (x, y)
double angle_deg = angle_rad * 180.0 / PI; // Radians to
Degrees!
double asin_val  = asin(s);            // arcsin (return
s [-π/2, π/2])
double acos_val  = acos(c);            // arccos (return
s [0, π])

// 6. Absolute & Hypotenuse:
double abs_val   = fabs(x);            // |x| for floati
ng point (fabsl for long double)
double hyp       = hypot(x, y);        // √(x^2 + y^2)
(safely prevents overflow)
```

Matrix I/O & Grid Helpers

`mat_io` | `O(N × M)`

Dynamic 2D vector input/output, char grids, and 4/8-direction movement arrays.

```
// 1. Numeric Matrix N x M
int n, m; cin >> n >> m;
vector<vector<int>> a(n, vector<int>(m));
for (int i = 0; i < n; i++)
  for (int j = 0; j < m; j++) cin >> a[i][j];

// Output Matrix (space-separated, newline per row)
for (int i = 0; i < n; i++) {
  for (int j = 0; j < m; j++) cout << a[i][j] << (j == m
- 1 ? "" : " ");
  cout << '\n';
}

// 2. Character Grid (Maze / Labyrinth input)
vector<string> grid(n);
for (int i = 0; i < n; i++) cin >> grid[i];

// 3. Direction Arrays for Grid Navigation
int dx4[] = {-1, 1, 0, 0}, dy4[] = {0, 0, -1, 1}; // Up,
Down, Left, Right (Rook / 4-step)
int dx8[] = {-1,-1,-1, 0, 0, 1, 1, 1};             // 8
directions (Queen / King / diagonals)
int dy8[] = {-1, 0, 1,-1, 1,-1, 0, 1};
// Knight moves (8 L-shaped jumps):
int knight_dx[] = {2, 2, 1, -1, 1, -1, -2, -2};
int knight_dy[] = {1, -1, 2, 2, -2, -2, -1, 1};
bool valid(int x, int y, int n, int m) { return x >= 0 &
& x < n && y >= 0 && y < m; }

// 4. Queen / Rook Ray-Casting Move (sliding until obsta
cle '#')
for (int i = 0; i < 4; i++) { // i < 4 for Rook, i < 8 f
or Queen (with dx8, dy8)
  int nx = x + dx[i], ny = y + dy[i];
  while (nx >= 0 && ny >= 0 && nx < n && ny < m) {
    if (grid[nx][ny] == '#') break;
    if (dis[nx][ny] == -1) {
      dis[nx][ny] = dis[x][y] + 1;
      q.push({nx, ny});
    }
    nx += dx[i]; ny += dy[i];
  }
}
```

All Type Converters & Parsers `convert`

Converting between strings, numbers, characters, bases, and splitting tokens.

```
// 1. Number <-> String
string s = to_string(12345LL);          // int / LL to string
int num = stoi("123");                   // string to int
long long lnum = stoll("1234567890123"); // string to long long
double d = stod("3.14159");              // string to double
long double ld = stold("3.1415926535");  // string to long double

// 2. Char <-> Digit & Alphabets
int digit = '7' - '0';                   // '7' -> 7
char ch = 7 + '0';                       // 7 -> '7'
int lower_idx = 'c' - 'a';               // 'c' -> 2 (0 to 25)
int upper_idx = 'C' - 'A';               // 'C' -> 2 (0 to 25)
char lower_c = tolower('A');             // 'A' -> 'a'
char upper_c = toupper('a');             // 'a' -> 'A'
// Entire string to lower/upper:
transform(s.begin(), s.end(), s.begin(), ::tolower);

// 3. Base Conversions (Binary, Hex, Octal)
int from_bin = stoi("10110", nullptr, 2);  // "10110" -> 22
int from_hex = stoi("1A", nullptr, 16);    // "1A" -> 26
int from_oct = stoi("75", nullptr, 8);     // "75" -> 61
string to_bin = bitset<32>(22).to_string(); // 22 -> "00...0010110"
to_bin = to_bin.substr(to_bin.find('1'));  // Trim leading zeros
stringstream ss; ss << hex << 255; string hex_str = ss.str(); // "ff"

// 4. String Tokenizer (split by space / delimiter)
stringstream ss2("apple banana cherry");
string word; while (ss2 >> word) { /* process each word */ }
stringstream ss3("1,2,3,4");
string item; while (getline(ss3, item, ',')) { int val = stoi(item); }
```

All STL Containers: Operations & Complexity `stl_containers`

Methods, syntax, and complexities for every standard competitive programming container.

```
// 1. VECTOR (O(1) push/pop_back, O(1) random access)
vector<int> v = {1, 2, 3};
v.push_back(4); v.pop_back(); v.front(); v.back(); v.size(); v.empty();
v.resize(10, 0); v.assign(5, -1); v.clear();
v.insert(v.begin() + 1, 99); v.erase(v.begin() + 1); // O(N)

// 2. PAIR & TUPLE
pair<int, string> p = {1, "ab"}; auto [id, name] = p; // C++17 structured binding
tuple<int, int, int> tp = {1, 2, 3}; auto [x, y, z] = tp; int f = get<0>(tp);

// 3. SET & MULTISET (Red-Black Tree - O(log N) operations, sorted)
set<int> s; s.insert(5); s.erase(5); bool has = s.count(5);
auto it_lb = s.lower_bound(x); // first element >= x
auto it_ub = s.upper_bound(x); // first element > x
multiset<int> ms; ms.insert(5); ms.insert(5);
ms.erase(ms.find(5)); // Erases ONLY ONE 5! (ms.erase(5) erases ALL 5s)

// 4. MAP (Sorted key-value - O(log N)) & Frequency Counter
map<string, int> mp; mp["apple"] = 5; mp.count("apple");
mp.erase("apple");
for (auto& [key, val] : mp) { /* sorted by key */ }
// Frequency counter snippet:
map<int, int> freq; for (int x : v) freq[x]++;
int max_f = 0, most_frequent = -1;
for (auto& [val, count] : freq) {
  if (count > max_f) { max_f = count; most_frequent = val; }
}

// 5. PRIORITY_QUEUE (Heap - O(log N) push/pop, O(1) top)
priority_queue<int> max_pq; // Max heap by default
priority_queue<int, vector<int>, greater<int>> min_pq; // Min heap
max_pq.push(10); max_pq.top(); max_pq.pop();

// 6. DEQUE, STACK & QUEUE (O(1) push/pop)
deque<int> dq; dq.push_front(1); dq.push_back(2); dq.pop_front(); dq.pop_back();
stack<int> st; st.push(1); st.top(); st.pop();
queue<int> q; q.push(1); q.front(); q.pop();

// 7. BITSET (Fixed-size bit array - 64x faster bitwise ops)
bitset<64> bs(15); bs.set(2); bs.reset(2); bs.flip(); bool bit = bs.test(0);
int set_bits = bs.count(); unsigned long long val = bs.to_ullong();
```

All STL Algorithms: Quick Syntax & Uses `stl_algo`

Essential functions in <algorithm> and <numeric> with parameter shortcuts.

```
// 1. Sorting & Bounds (v is vector<int>)
sort(all(v));                               // Ascending: O(N log N)
sort(rall(v));                              // Descending: O(N log N)
stable_sort(all(v), comp);                  // Preserves relative order
auto it1 = lower_bound(all(v), x);          // First iterator with *it >= x
int idx1 = lower_bound(all(v), x) - v.begin(); // 0-based index of first >= x
auto it2 = upper_bound(all(v), x);          // First iterator with *it > x
bool found = binary_search(all(v), x);      // True if x exists (O(log N))

// 2. Element Operations & Deduplication
v.erase(unique(all(v)), v.end());           // Remove duplicates (sort first!)
reverse(all(v));                            // Invert order: O(N)
rotate(v.begin(), v.begin() + k, v.end()); // Left-rotates by k positions
int mn = *min_element(all(v));              // Minimum element: O(N)
int mx = *max_element(all(v));              // Maximum element: O(N)
int cnt = count(all(v), x);                 // Count occurrences of x: O(N)
fill(all(v), 0);                            // Fill all with 0

// 3. Numeric & Prefix Sums (<numeric>)
int sum = accumulate(all(v), 0LL);          // Sum (use 0LL to prevent overflow!)
vector<int> pref(n);
partial_sum(all(v), pref.begin());          // Prefix sum array in-place: O(N)
iota(all(v), 0);                            // Fills with 0, 1, 2, ..., n-1

// 4. Permutations & Order Statistics
next_permutation(all(v));                   // Next lexicographical order
prev_permutation(all(v));                   // Previous order
// nth_element: Rearranges so element at index k is in its sorted position: O(N) avg
nth_element(v.begin(), v.begin() + k, v.end()); // v[k] is the k-th smallest!
```

Coordinate Compression (O(N log N)) `compress` | O(N log N)

Maps large arbitrary coordinates (up to 10^9) to dense ranks [0, K-1].

```
// 1. In-place coordinate compression on vector 'a':
vector<int> vals = a;
sort(vals.begin(), vals.end());
vals.erase(unique(vals.begin(), vals.end()), vals.end());
for (int &x : a) {
  x = lower_bound(vals.begin(), vals.end(), x) - vals.begin(); // 0-based rank
}
// 2. Query original value from compressed index: vals[compressed_id]
```

3. MATH, NUMBER THEORY & BITS

Greatest Common Divisor (GCD) `gcdf` | O(log(min(a,b)))

```
int gcd(int a, int b) {
  while (b) { a %= b; swap(a, b); }
  return a;
}
```

Least Common Multiple (LCM) `lcmf` | O(log(min(a,b)))

```
int lcm(int a, int b) {
  if (a == 0 || b == 0) return 0;
  return a / __gcd(a, b) * b;
}
```

Sieve of Eratosthenes `gprime` | O(N log log N)

```
vector<int> primes; vector<bool> is_prime;
void gprime(int n) {
  is_prime.assign(n + 1, true);
  is_prime[0] = is_prime[1] = false;
  for (int i = 2; i * i <= n; i++)
    if (is_prime[i])
      for (int j = i * i; j <= n; j += i) is_prime[j] = false;
  primes.clear();
  for (int i = 2; i <= n; i++) if (is_prime[i]) primes.push_back(i);
}
```

Linear Sieve & Fast Factorization (SPF) `spf` | O(N) + O(log X)

Precomputes Smallest Prime Factor (SPF) in O(N). Factorizes Q queries in O(log X).

```
const int MAXV = 1e7 + 5;
vector<int> spf(MAXV), primes_list;
void linear_sieve() {
  for (int i = 2; i < MAXV; ++i) {
    if (spf[i] == 0) { spf[i] = i; primes_list.push_back(i); }
    for (int p : primes_list) {
      if (p > spf[i] || i * p >= MAXV) break;
      spf[i * p] = p;
    }
  }
}
// Factorize any number x in O(log x):
vector<pair<int, int>> get_factors(int x) {
  vector<pair<int, int>> factors;
  while (x > 1) {
    int p = spf[x], count = 0;
    while (x % p == 0) { count++; x /= p; }
    factors.push_back({p, count});
  }
  return factors;
}
```

Binary Exponentiation with Modulo `binExpoMod` | O(log b)

```
long long binpow(long long a, long long b, long long m)
{
  a %= m; long long res = 1;
  while (b > 0) {
    if (b & 1) res = res * a % m;
    a = a * a % m; b >>= 1;
  }
  return res;
}
```

Binary Exponentiation (Standard) `binExpo` | O(log b)

```
long long binpow(long long a, long long b) {
  long long res = 1;
  while (b > 0) {
    if (b & 1) res = res * a;
    a = a * a; b >>= 1;
  }
  return res;
}
```

Count Set Bits (1s) from 0 to N `numofonef` | O(log N)

The total may exceed 64 bits; display it with the Section 1.4 print128 helper.

```
__uint128_t numofone(unsigned long long n) {
  __uint128_t total = 0, count = (__uint128_t)n + 1;
  for (int bit = 0; bit < 64; ++bit) {
    __uint128_t half = (__uint128_t)1 << bit;
    __uint128_t period = half << 1;
    __uint128_t rem = count % period;
    total += (count / period) * half;
    if (rem > half) total += rem - half;
  }
  return total;
}
```

Extended Euclidean & Modular Inverse `O(log(min(a,b)))`

An inverse exists only when gcd(a,m)=1; requires m>1.

```
int ext_gcd(int a, int b, int &x, int &y) {
  if (b == 0) { x = 1; y = 0; return a; }
  int x1, y1, g = ext_gcd(b, a % b, x1, y1);
  x = y1; y = x1 - (a / b) * y1;
  return g;
}
int mod_inv(int a, int m) {
  assert(m > 1);
  a %= m; if (a < 0) a += m;
  int x, y, g = ext_gcd(a, m, x, y);
  assert(g == 1);
  x %= m; if (x < 0) x += m;
  return x;
}
```

Combinatorics: nCr in O(R)

```
int nCr(int n, int r) {
  if (r < 0 || r > n) return 0;
  if (r > n / 2) r = n - r;
  int ans = 1;
  for (int i = 1; i <= r; ++i) ans = ans * (n - i + 1) / i;
  return ans;
}
```

Stable Quadratic Equation Solver `quadf` | `O(1)`

Citardauq formula avoids catastrophic cancellation when b ≈ √D. Handles a=0 linear, D<0, D=0.

```
using ld = long double;
const ld EPS = 1e-12;

void solve_quadratic(ld a, ld b, ld c) {
  cout << fixed << setprecision(12);
  if (fabsl(a) < EPS) {
    if (fabsl(b) < EPS) cout << "No solution\n";
    else cout << -c / b << '\n';
    return;
  }
  ld d = b * b - 4 * a * c;
  if (d < -EPS) { cout << "No real roots\n"; return; }
  if (fabsl(d) <= EPS) { cout << -b / (2 * a) << '\n'; return; }
  ld sq = sqrtl(d);
  ld q = -0.5L * (b + copysignl(sq, b)); // Numerically stable
  ld x1 = q / a, x2 = c / q;
  if (x1 > x2) swap(x1, x2);
  cout << x1 << '\n' << x2 << '\n';
}
```

Basis Conversion (Base 2 to 16) `base_conv` | `O(log_b N)`

Bidirectional conversion between base-b string and unsigned long long (b ≤ 16).

```
const string DIGITS = "0123456789ABCDEF";

// Base-b string to unsigned long long
unsigned long long basis_string_to_number(string &s, int b) {
  unsigned long long result = 0ULL;
  for (char d : s) {
    result = b * result + (find(DIGITS.begin(), DIGITS.end(), d) - DIGITS.begin());
  }
  return result;
}

// Unsigned long long to base-b string
string number_to_basis_string(unsigned long long n, int b) {
  vector<char> ds;
  do {
    ds.push_back(DIGITS[n % b]);
    n /= b;
  } while (n != 0);
  return string(ds.rbegin(), ds.rend());
}
```

All Binary Operations & Bit Tricks `bits` | `O(1)`

GCC builtins, essential bitwise manipulations, power of 2, and submask iteration.

```
// 1. GCC Builtins (Append 'll' for 64-bit: e.g. __builtin_popcountll)
int ones = __builtin_popcountll(x);   // Count set bits (number of 1s)
int clz  = __builtin_clzll(x);        // Count leading zeros (undefined if x=0!)
int ctz  = __builtin_ctzll(x);        // Count trailing zeros (lowest set bit index)
int par  = __builtin_parityll(x);     // Parity: 1 if odd number of 1s, 0 if even
int msb  = 63 - __builtin_clzll(x);   // Index of highest set bit (= floor(log2(x)))

// 2. Essential Bit Hacks
bool is_set   = (n >> k) & 1;          // Check if k-th bit is 1 (0-indexed)
n |= (1LL << k);                      // Set k-th bit to 1
n &= ~(1LL << k);                     // Clear k-th bit to 0
n ^= (1LL << k);                      // Toggle k-th bit (flip 0 <-> 1)
bool is_pow2  = n > 0 && !(n & (n - 1)); // True if n is a power of 2
int lsb       = n & (-n);             // Isolates lowest set bit: e.g. 12 (1100) -> 4 (0100)
n &= (n - 1);                         // Clears the lowest set bit

// 3. Iterate All Submasks of Mask m (descending order):
for (int s = m; s > 0; s = (s - 1) & m) { /* s is a non-empty submask */ }
// 4. Iterate All Masks of Size N:
for (int mask = 0; mask < (1 << n); ++mask) { /* O(2^N) */ }
```

Complete Modulo Sheet & O(1) nCr `mod_sheet`

Normalizes negative inputs. Fermat inverse needs a nonzero residue; factorial nCr needs prime MOD and n < MOD.

```
const int MOD = 1e9 + 7; // or 998244353

// 1. Safe Operations (handles negatives & prevents over
flow)
long long norm(long long x) { x %= MOD; return x < 0 ? x
+ MOD : x; }
long long add(long long a, long long b) { return (norm
(a) + norm(b)) % MOD; }
long long sub(long long a, long long b) { return (norm
(a) - norm(b) + MOD) % MOD; }
long long mul(long long a, long long b) { return norm(a)
* norm(b) % MOD; }

// 2. Fast Exponentiation: (base^exp) % MOD
long long power(long long base, long long exp) {
  long long res = 1; base = norm(base);
  while (exp > 0) {
    if (exp % 2 == 1) res = (res * base) % MOD;
    base = (base * base) % MOD;
    exp /= 2;
  }
  return res;
}

// 3. Modular Inverse via Fermat (MOD must be prime) & D
ivision
long long modInverse(long long n) { assert(norm(n) !=
0); return power(n, MOD - 2); }
long long divi(long long a, long long b) { return mul(a,
modInverse(b)); }

// 4. O(N) Precomputation for O(1) nCr Modulo P (N up to
10^6):
const int MAXN = 1000005;
long long fact[MAXN], invFact[MAXN];
void precompute_nCr(int n = MAXN - 1) {
  assert(0 <= n && n < MOD);
  fact[0] = invFact[0] = 1;
  for (int i = 1; i <= n; i++) fact[i] = mul(fact[i -
1], i);
  invFact[n] = modInverse(fact[n]);
  for (int i = n - 1; i >= 1; i--) invFact[i] = mul(invF
act[i + 1], i + 1);
}
long long nCr_mod(int n, int r) {
  if (r < 0 || r > n) return 0;
  return mul(fact[n], mul(invFact[r], invFact[n - r]));
}
```

4. GRAPH TRAVERSALS & INPUTS

Adjacency List Input `adjlist`

```
int V, E; cin >> V >> E;
vector<vector<pair<int, int>>> adj(V);
for (int i = 0; i < E; ++i) {
  int u, v, w; cin >> u >> v >> w;
  adj[u].push_back({v, w}); // adj[v].push_back({u, w});
// if undirected
}
```

Edge List Input `adjmat`

```
int V, E; cin >> V >> E;
vector<vector<int>> edges;
for (int i = 0; i < E; ++i) {
  int u, v, w; cin >> u >> v >> w; edges.push_back({u,
v, w});
}
```

DFS Traversal with Graph Input `dfs` | `O(V + E)`

```
void dfs(int node, const vector<int> adj[], vector<bool>
& visited) {
  visited[node] = true;
  for (int neighbor : adj[node])
    if (!visited[neighbor]) dfs(neighbor, adj, visited);
}
signed main() {
  int n, m; cin >> n >> m;
  vector<int> adj[n + 1]; vector<bool> visited(n + 1, fa
lse);
  for (int i = 0; i < m; ++i) {
    int u, v; cin >> u >> v;
    adj[u].push_back(v); adj[v].push_back(u);
  }
  for (int i = 1; i <= n; ++i) if (!visited[i]) dfs(i, a
dj, visited);
  return 0;
}
```

DFS Function Only (Standalone) `dfsfunc`

```
void dfs(int node, const vector<int> adj[], vector<bool>
& visited) {
  visited[node] = true;
  for (int neighbor : adj[node])
    if (!visited[neighbor]) dfs(neighbor, adj, visited);
}
```

BFS Traversal with Graph Input `bfs` | `O(V + E)`

```
void bfs(int start, const vector<int> adj[], vector<bool
>& visited) {
  queue<int> q; q.push(start); visited[start] = true;
  while (!q.empty()) {
    int node = q.front(); q.pop();
    for (int neighbor : adj[node]) {
      if (!visited[neighbor]) { visited[neighbor] = tru
e; q.push(neighbor); }
    }
  }
}
signed main() {
  int n, m; cin >> n >> m;
  vector<int> adj[n + 1]; vector<bool> visited(n + 1, fa
lse);
  for (int i = 0; i < m; ++i) {
    int u, v; cin >> u >> v;
    adj[u].push_back(v); adj[v].push_back(u);
  }
  for (int i = 1; i <= n; ++i) if (!visited[i]) bfs(i, a
dj, visited);
}
```

BFS Function Only (Standalone) `bfsfunc`

```
void bfs(int start, const vector<int> adj[], vector<bool
>& visited) {
  queue<int> q; q.push(start); visited[start] = true;
  while (!q.empty()) {
    int node = q.front(); q.pop();
    for (int neighbor : adj[node]) {
      if (!visited[neighbor]) { visited[neighbor] = tru
e; q.push(neighbor); }
    }
  }
}
```

2D Grid BFS (Maze Shortest Path & Queen/Knight Moves) `bfs_grid` | O(N × M)

Shortest path in a character grid from 'S' to 'E' with obstacles '#', supporting 4-step, Knight, and Queen/Rook sliding moves.

```
int main() {
  int n, m; cin >> n >> m;
  vector<string> grid(n);
  for (int i = 0; i < n; i++) cin >> grid[i];

  vector<vector<int>> dis(n, vector<int>(m, -1));
  queue<pair<int,int>> q;
  int sx, sy, ex, ey;

  for (int i = 0; i < n; i++) {
    for (int j = 0; j < m; j++) {
      if (grid[i][j] == 'S') { sx = i; sy = j; }
      if (grid[i][j] == 'E') { ex = i; ey = j; }
    }
  }

  q.push({sx, sy}); dis[sx][sy] = 0;
  int dx[] = {-1, 1, 0, 0}, dy[] = {0, 0, -1, 1}; // 4-d
irectional step

  while (!q.empty()) {
    auto [x, y] = q.front(); q.pop();
    if (x == ex && y == ey) {
      cout << "Shortest distance = " << dis[x][y] <<
'\n';
      return 0;
    }

    // 1. Single-Step Move (4 directions):
    for (int i = 0; i < 4; i++) {
      int nx = x + dx[i], ny = y + dy[i];
      if (nx >= 0 && nx < n && ny >= 0 && ny < m &&
          grid[nx][ny] != '#' && dis[nx][ny] == -1) {
        q.push({nx, ny}); dis[nx][ny] = dis[x][y] + 1;
      }
    }

    // 2. Queen / Rook Move (Ray-casting / sliding until
'#'):
    /*
    for (int i = 0; i < 4; i++) { // i < 4 for Rook, i <
8 for Queen
      int nx = x + dx[i], ny = y + dy[i];
      while (nx >= 0 && ny >= 0 && nx < n && ny < m) {
        if (grid[nx][ny] == '#') break;
        if (dis[nx][ny] == -1) {
          dis[nx][ny] = dis[x][y] + 1;
          q.push({nx, ny});
        }
        nx += dx[i]; ny += dy[i];
      }
    }
    */
  }
  cout << "Destination not reachable\n";
}
```

Queen / Rook Move (Grid Ray-Casting BFS) `queen_move` | O(N × M)

Sliding piece move across empty cells in 4 directions (Rook) or 8 directions (Queen) until hitting obstacle '#' or grid boundary.

```
// Queen / Rook ray-casting move (i < 4 for Rook, i < 8
for Queen):
for (int i = 0; i < 4; i++) {
  int nx = x + dx[i];
  int ny = y + dy[i];
  while (nx >= 0 && ny >= 0 && nx < n && ny < m) {
    if (grid[nx][ny] == '#') break;
    if (dis[nx][ny] == -1) {
      dis[nx][ny] = dis[x][y] + 1;
      q.push({nx, ny});
    }
    nx += dx[i];
    ny += dy[i];
  }
}
```

BFS Traversal with Path Printing (Predecessors) `bfs_path` | O(V + E)

BFS traversal using 3-color state (white/grey/black), distance array, predecessor tracking, and path reconstruction.

```
void bfs(vector<vector<int>>& adj_list, int src, int n,
int destination) {
  vector<string> color(n, "white");
  vector<int> dis(n, INT_MAX), pred(n, -1);
  queue<int> q;
  q.push(src); color[src] = "grey"; dis[src] = 0;

  while (!q.empty()) {
    int par = q.front(); q.pop(); color[par] = "black";
    for (auto child : adj_list[par]) {
      if (color[child] == "white") {
        q.push(child); color[child] = "grey";
        dis[child] = dis[par] + 1; pred[child] = par;
      }
    }
  }

  int node = destination;
  vector<int> path;
  while (node != -1) { path.push_back(node); node = pred
[node]; }
  reverse(path.begin(), path.end());
  for (int v : path) cout << v << " ";
  cout << '\n';

  if (dis[destination] == INT_MAX) cout << "disconnected
\n";
  else cout << "distance-> " << dis[destination] <<
'\n';
}
```

Topological Sort (Kahn's BFS) O(V + E)

```
vector<int> topo_sort(int n, const vector<vector<int>> &
adj) {
  vector<int> in_deg(n, 0), order;
  for (int u = 0; u < n; ++u)
    for (int v : adj[u]) in_deg[v]++;
  queue<int> q;
  for (int i = 0; i < n; ++i) if (in_deg[i] == 0) q.push
(i);
  while (!q.empty()) {
    int u = q.front(); q.pop(); order.push_back(u);
    for (int v : adj[u]) if (--in_deg[v] == 0) q.push
(v);
  }
  return (order.size() == n) ? order : vector<int>();
}
```

Bipartite Graph Check & 2-Coloring (BFS) `bipartite` | O(V + E)

Returns false if an odd-length cycle exists. Stores colors {0, 1} in vector.

```
bool is_bipartite(int n, const vector<vector<int>> &adj,
vector<int> &color) {
  color.assign(n + 1, -1);
  for (int start = 1; start <= n; ++start) {
    if (color[start] != -1) continue;
    queue<int> q; q.push(start); color[start] = 0;
    while (!q.empty()) {
      int u = q.front(); q.pop();
      for (int v : adj[u]) {
        if (color[v] == -1) {
          color[v] = color[u] ^ 1; q.push(v);
        } else if (color[v] == color[u]) return false;
// Odd cycle detected
      }
    }
  }
  return true;
}
```

5. SHORTEST PATHS & CYCLES

Dijkstra (PQ + Path Print) `dijkstra` | O((V+E)log V)

```
int n, m; cin >> n >> m;
vector<vector<pair<int, int>>> adj(n + 1);
for (int i = 0; i < m; i++) {
  int u, v, w; cin >> u >> v >> w;
  adj[u].emplace_back(v, w); adj[v].emplace_back(u, w);
}
int s = 1, inf = LLONG_MAX / 2;
vector<int> dis(n + 1, inf), parent(n + 1, -1);
dis[s] = 0;
priority_queue<pair<int, int>, vector<pair<int, int>>, g
reater<pair<int, int>>> pq;
pq.emplace(0, s);
while (!pq.empty()) {
  auto [dv, v] = pq.top(); pq.pop();
  if (dv > dis[v]) continue;
  for (auto &edge : adj[v]) {
    int to = edge.first, d = edge.second;
    if (dis[v] + d < dis[to]) {
      dis[to] = dis[v] + d; parent[to] = v; pq.emplace(d
is[to], to);
    }
  }
}
if (dis[n] == inf) cout << -1 << endl;
else {
  vector<int> path;
  for (int i = n; i != -1; i = parent[i]) path.push_back
(i);
  reverse(path.begin(), path.end());
  for (int node : path) cout << node << " ";
  cout << endl;
}
```

Dijkstra Algorithm (Set-Based) `dijkstraSet` | O((V+E)log V)

```
const long long INF = LLONG_MAX / 4;
vector<vector<pair<int, long long>>> adj;

void dijkstra(int s, vector<long long>& d, vector<int>&
p) {
    int n = adj.size();
    d.assign(n, INF);
    p.assign(n, -1);
    d[s] = 0;
    set<pair<long long, int>> q;
    q.insert({0, s});
    while (!q.empty()) {
        int v = q.begin()->second;
        q.erase(q.begin());

        for (auto [to, len] : adj[v]) {
            assert(len >= 0);
            if (d[v] + len < d[to]) {
                q.erase({d[to], to});
                d[to] = d[v] + len;
                p[to] = v;
                q.insert({d[to], to});
            }
        }
    }
}
```

0-1 BFS (Deque Shortest Path) `bfs01` | O(V + E)

Fast shortest path on graphs where edge weights are only 0 or 1.

```
vector<int> zero_one_bfs(int start, int n, const vector<
vector<pair<int, int>>> &adj) {
  vector<int> dist(n + 1, 1e9); deque<int> dq;
  dist[start] = 0; dq.push_back(start);
  while (!dq.empty()) {
    int u = dq.front(); dq.pop_front();
    for (auto &[v, w] : adj[u]) {
      if (dist[u] + w < dist[v]) {
        dist[v] = dist[u] + w;
        if (w == 0) dq.push_front(v); // Push 0-weight t
o front!
        else dq.push_back(v);        // Push 1-weight to
back!
      }
    }
  }
  return dist;
}
```

Bellman-Ford Algorithm `bellman_ford` | O(V * E)

```
const long long BF_INF = LLONG_MAX / 4;
pair<bool, vector<long long>> bellman_ford(
    int n, const vector<tuple<int,int,long long>>& edge
s, int S) {
    vector<long long> dis(n, BF_INF);
    dis[S] = 0;
    for (int i = 0; i < n - 1; i++) {
        bool updated = false;
        for (auto [u,v,w] : edges) {
            if (dis[u] != BF_INF && dis[u] + w < dis[v])
{
                dis[v] = dis[u] + w;
                updated = true;
            }
        }
        if (!updated) break;
    }
    for (auto [u,v,w] : edges)
        if (dis[u] != BF_INF && dis[u] + w < dis[v]) ret
urn {true, dis};
    return {false, dis};
}
```

Negative Cycle Detection & Path Print `negcyclef` | O(V * E)

```
int n, m; cin >> n >> m;
vector<vector<int>> edges;
for (int i = 0; i < m; ++i) {
  int u, v, w; cin >> u >> v >> w; edges.push_back({u,
v, w});
}
vector<int> dist(n + 1, 0), p(n + 1, -1);
int x;
for (int i = 0; i < n; i++) {
  x = -1;
  for (auto &edge : edges) {
    int u = edge[0], v = edge[1], w = edge[2];
    if (dist[u] + w < dist[v]) {
      dist[v] = dist[u] + w; p[v] = u; x = v;
    }
  }
}
if (x == -1) { cout << "NO" << endl; return 0; }
for (int i = 0; i < n; i++) x = p[x];
cout << "YES" << endl;
vector<int> path; int cur = x; path.push_back(cur);
while (true) {
  cur = p[cur]; path.push_back(cur);
  if (cur == x) break;
}
reverse(all(path));
for (int node : path) cout << node << " ";
cout << endl;
```

Floyd-Warshall (All-Pairs) O(V^3)

```
for (int k = 0; k < n; ++k)
  for (int i = 0; i < n; ++i)
    for (int j = 0; j < n; ++j)
      if (dist[i][k] < INF && dist[k][j] < INF)
        dist[i][j] = min(dist[i][j], dist[i][k] + dist
[k][j]);
```

6. TREE ALGORITHMS

Tree Diameter via Double BFS `treeDiameter` | O(V + E)

```
pair<int, int> bfs_farthest(int start, const vector<vect
or<int>>& adj, int n) {
  vector<int> dist(n + 1, -1);
  queue<int> q; q.push(start); dist[start] = 0;
  int farthest_node = start;
  while (!q.empty()) {
    int node = q.front(); q.pop();
    for (int neighbor : adj[node]) {
      if (dist[neighbor] == -1) {
        dist[neighbor] = dist[node] + 1; q.push(neighbo
r);
        if (dist[neighbor] > dist[farthest_node]) farthe
st_node = neighbor;
      }
    }
  }
  return {farthest_node, dist[farthest_node]};
}
signed main() {
  fastIO(); int n; cin >> n;
  vector<vector<int>> adj(n + 1);
  for (int i = 0; i < n - 1; ++i) {
    int u, v; cin >> u >> v; adj[u].push_back(v); adj
[v].push_back(u);
  }
  int far1 = bfs_farthest(1, adj, n).first;
  int diameter = bfs_farthest(far1, adj, n).second;
  cout << diameter << '\n';
}
```

Lowest Common Ancestor (Binary Lifting) `lca` | O(N log N) + O(log N)

Tree binary lifting for LCA, k-th ancestor, and node distance queries.

```
const int MAXN = 2e5 + 5, LOGN = 20;
vector<int> tree_adj[MAXN];
int up[MAXN][LOGN], depth[MAXN];

void build_lca(int root) { // iterative: safe for a long
chain
    vector<int> q = {root};
    for (int i = 0; i < LOGN; ++i) up[root][i] = -1;
    depth[root] = 0;
    for (size_t head = 0; head < q.size(); ++head) {
        int u = q[head];
        for (int v : tree_adj[u]) if (v != up[u][0]) {
            depth[v] = depth[u] + 1;
            up[v][0] = u;
            for (int i = 1; i < LOGN; ++i)
                up[v][i] = up[v][i-1] == -1 ? -1 : up[up
[v][i-1]][i-1];
            q.push_back(v);
        }
    }
}

int get_lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    int diff = depth[u] - depth[v];
    for (int i = 0; i < LOGN; ++i)
        if ((diff >> i) & 1) u = up[u][i];
    if (u == v) return u;
    for (int i = LOGN - 1; i >= 0; --i) {
        if (up[u][i] != up[v][i]) { u = up[u][i]; v = up
[v][i]; }
    }
    return up[u][0];
}

int tree_dist(int u, int v) {
    return depth[u] + depth[v] - 2 * depth[get_lca(u,
v)];
}
```

7. DATA STRUCTURES (DSU & RANGE TREES)

Disjoint Set Union (Union by Size) `dsu-size` | O(α(N)) ≈ O(1)

```
class DisjointSet {
  vector<int> rank, parent, size;
public:
  DisjointSet(int n) {
    rank.resize(n + 1, 0); parent.resize(n + 1); size.re
size(n + 1, 1);
    for (int i = 0; i <= n; i++) parent[i] = i;
  }
  int findUPar(int node) {
    if (node == parent[node]) return node;
    return parent[node] = findUPar(parent[node]);
  }
  void unionBySize(int u, int v) {
    int ulpU = findUPar(u), ulpV = findUPar(v);
    if (ulpU == ulpV) return;
    if (size[ulpU] < size[ulpV]) {
      parent[ulpU] = ulpV; size[ulpV] += size[ulpU];
    } else {
      parent[ulpV] = ulpU; size[ulpU] += size[ulpV];
    }
  }
  bool connected(int u, int v) { return findUPar(u) == f
indUPar(v); }
};
```

Fenwick Tree (BIT: Sum & Range Average) O(log N)

Point update, range sum, integer & decimal range average (`avg = sum / count`).

```
struct Fenwick {
  int n; vector<int> bit;
  Fenwick(int n) : n(n), bit(n + 1, 0) {}
  void add(int idx, int val) {
    for (; idx <= n; idx += idx & -idx) bit[idx] += val;
  }
  int query(int idx) {
    int sum = 0;
    for (; idx > 0; idx -= idx & -idx) sum += bit[idx];
    return sum;
  }
  int range_query(int l, int r) { return query(r) - quer
y(l - 1); }
  // Average = sum / number_of_elements (count = r - l +
1)
  int avg(int l, int r) { return range_query(l, r) / (r
- l + 1); } // integer
  double avg_d(int l, int r) { return (double)range_quer
y(l, r) / (r - l + 1); } // decimal
};
```

Iterative Segment Tree (Range Sum) O(log N)

```
struct SegTree {
  int n; vector<int> t;
  SegTree(int n) : n(n), t(2 * n, 0) {}
  void update(int p, int val) {
    for (t[p += n] = val; p > 1; p >>= 1) t[p >> 1] = t
[p] + t[p ^ 1];
  }
  int query(int l, int r) { // sum on [l, r]
    int res = 0;
    for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
      if (l & 1) res += t[l++];
      if (r & 1) res += t[--r];
    }
    return res;
  }
};
```

Segment Tree for MIN & MAX (Minimal Changes) `O(log N)`

Minimal changes to adapt base SegTree for Range Min / Max Query (RMQ).

```
// === SEGMENT TREE FOR MIN ===
// 1. Initialize with infinity:
SegTree(int n) : n(n), t(2 * n, numeric_limits<int>::max
()) {}
// 2. Update: change t[p>>1] = t[p] + t[p^1] to:
void update(int p, int val) {
  for (t[p += n] = val; p > 1; p >>= 1) t[p >> 1] = min
(t[p], t[p ^ 1]);
}
// 3. Query min on [l, r]:
int query(int l, int r) {
  int res = numeric_limits<int>::max();
  for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
    if (l & 1) res = min(res, t[l++]);
    if (r & 1) res = min(res, t[--r]);
  }
  return res;
}

// === SEGMENT TREE FOR MAX ===
// 1. Initialize: SegTree(int n) : n(n), t(2 * n, numeri
c_limits<int>::lowest()) {}
// 2. Update:     t[p >> 1] = max(t[p], t[p ^ 1]);
// 3. Query:      res = numeric_limits<int>::lowest();
//                if (l & 1) res = max(res, t[l++]);
//                if (r & 1) res = max(res, t[--r]);
```

KACTL Segment Tree (Associative f, [b, e)) `SegTreeKACTL` | `O(log N)`

Zero-indexed, [b, e) semi-open interval. Modify T, f, unit for min/sum/gcd.

```
struct Tree {
  typedef int T;
  static constexpr T unit = numeric_limits<T>::lowest();
  T f(T a, T b) { return max(a, b); } // any associative
fn
  vector<T> s; int n;
  Tree(int n = 0, T def = unit) : s(2 * n, def), n(n) {}
  void update(int pos, T val) {
    for (s[pos += n] = val; pos /= 2;)
      s[pos] = f(s[pos * 2], s[pos * 2 + 1]);
  }
  T query(int b, int e) { // query [b, e)
    T ra = unit, rb = unit;
    for (b += n, e += n; b < e; b /= 2, e /= 2) {
      if (b % 2) ra = f(ra, s[b++]);
      if (e % 2) rb = f(s[--e], rb);
    }
    return f(ra, rb);
  }
};
```

Lazy Segment Tree (Range Add, Set & Max) `LazySegTree` | `O(log N)`

Supports range add, range set, and range max queries on [L, R).

```
struct LazySegTree {
    struct Node { long long mx=0, add=0, set=0; bool has
_set=false; };
    int n; vector<Node> t;
    explicit LazySegTree(const vector<long long>& a) : n
(a.size()), t(4 * max(1, n)) {
        if (n) build(1, 0, n, a);
    }
    void build(int p, int l, int r, const vector<long lo
ng>& a) {
        if (r-l==1) { t[p].mx=a[l]; return; }
        int m=(l+r)/2; build(2*p,l,m,a); build(2*p+1,m,
r,a);
        t[p].mx=max(t[2*p].mx,t[2*p+1].mx);
    }
    void apply_set(int p, long long x) {
        t[p].mx=x; t[p].set=x; t[p].add=0; t[p].has_set=
true;
    }
    void apply_add(int p, long long x) {
        t[p].mx+=x;
        if (t[p].has_set) t[p].set+=x;
        else t[p].add+=x;
    }
    void push(int p) {
        if (t[p].has_set) {
            apply_set(2*p,t[p].set); apply_set(2*p+1,t
[p].set);
            t[p].has_set=false;
        }
        if (t[p].add) {
            apply_add(2*p,t[p].add); apply_add(2*p+1,t
[p].add);
            t[p].add=0;
        }
    }
    void update(int p,int l,int r,int L,int R,long long
x,bool assign) {
        if (R<=l || r<=L) return;
        if (L<=l && r<=R) { if (assign) apply_set(p,x);
else apply_add(p,x); return; }
        push(p); int m=(l+r)/2;
        update(2*p,l,m,L,R,x,assign); update(2*p+1,m,r,
L,R,x,assign);
        t[p].mx=max(t[2*p].mx,t[2*p+1].mx);
    }
    long long get(int p,int l,int r,int L,int R) {
        if (R<=l || r<=L) return LLONG_MIN;
        if (L<=l && r<=R) return t[p].mx;
        push(p); int m=(l+r)/2;
        return max(get(2*p,l,m,L,R),get(2*p+1,m,r,L,R));
    }
    void set(int L,int R,long long x) { assert(0<=L && L
<=R && R<=n); if(L<R) update(1,0,n,L,R,x,true); }
    void add(int L,int R,long long x) { assert(0<=L && L
<=R && R<=n); if(L<R) update(1,0,n,L,R,x,false); }
    long long query(int L,int R) { assert(0<=L && L<R &&
R<=n); return get(1,0,n,L,R); }
};
```

Sparse Table (O(1) Static RMQ / GCD) `sparsetable` | `O(N log N) + O(1)`

Idempotent range queries (min, max, gcd) in O(1) time without tree overhead.

```
template<typename T>
struct SparseTable {
  int n, K;
  vector<vector<T>> st;
  SparseTable(const vector<T>& a) {
    n = a.size(); K = __lg(n) + 1;
    st.assign(K, vector<T>(n));
    st[0] = a;
    for (int j = 1; j < K; ++j)
      for (int i = 0; i + (1 << j) <= n; ++i)
        st[j][i] = min(st[j - 1][i], st[j - 1][i + (1 <<
(j - 1))]);
  }
  T query(int L, int R) { // 0-based inclusive range [L,
R]
    int j = __lg(R - L + 1);
    return min(st[j][L], st[j][R - (1 << j) + 1]);
  }
};
```

8. STRING ALGORITHMS (KMP & TRIE)

Knuth-Morris-Pratt (KMP) Algorithm `kmp` | O(N + M)

```
vector<int> buildLPS(const string &pat) {
  int m = pat.size(), len = 0;
  vector<int> lps(m, 0);
  for (int i = 1; i < m;) {
    if (pat[i] == pat[len]) { len++; lps[i] = len; i++;
}
    else {
      if (len != 0) len = lps[len - 1];
      else { lps[i] = 0; i++; }
    }
  }
  return lps;
}
vector<int> KMPsearch(const string &text, const string &
pat) {
  int n = text.size(), m = pat.size();
  vector<int> lps = buildLPS(pat), positions;
  int i = 0, j = 0;
  while (i < n) {
    if (text[i] == pat[j]) {
      i++; j++;
      if (j == m) { positions.push_back(i - j); j = lps
[j - 1]; }
    } else {
      if (j != 0) j = lps[j - 1];
      else i++;
    }
  }
  return positions;
}
```

Trie (Prefix Tree) & Binary Trie (Max XOR) `trie` | O(|S|) / O(30×N)

Prefix word dictionary and 32-bit binary trie to find maximum XOR pair.

```
// 1. Character Trie (Prefix search & string frequency)
struct TrieNode { int next[26] = {}, pass = 0, end = 0;
};
vector<TrieNode> trie(1);
void insert_str(const string &s) {
  int cur = 0;
  for (char ch : s) {
    int c = ch - 'a';
    if (!trie[cur].next[c]) {
      trie[cur].next[c] = trie.size();
      trie.emplace_back();
    }
    cur = trie[cur].next[c]; trie[cur].pass++;
  }
  trie[cur].end++;
}

// 2. Binary Trie (Maximum XOR of two numbers in array)
struct BitTrie { int nxt[2] = {}; };
vector<BitTrie> bt(1);
void insert_num(int x) {
  int cur = 0;
  for (int i = 30; i >= 0; --i) {
    int b = (x >> i) & 1;
    if (!bt[cur].nxt[b]) { bt[cur].nxt[b] = bt.size(); b
t.emplace_back(); }
    cur = bt[cur].nxt[b];
  }
}
int query_max_xor(int x) {
  int cur = 0, max_xor = 0;
  for (int i = 30; i >= 0; --i) {
    int b = (x >> i) & 1, opp = b ^ 1;
    if (bt[cur].nxt[opp]) { max_xor |= (1 << i); cur = b
t[cur].nxt[opp]; }
    else cur = bt[cur].nxt[b];
  }
  return max_xor;
}
```

9. PARADIGMS & DYNAMIC PROGRAMMING

Bitmasking Search (Apple Division) `bitMaskingAppleDivision`

```
int n; cin >> n;
vector<int> v(n);
for (int &x : v) cin >> x;
int ans = LLONG_MAX;
for (int i = 0; i < (1 << n); i++) {
  int a = 0, b = 0;
  for (int pos = 0; pos < n; pos++) {
    if (i & (1 << pos)) a += v[pos];
    else b += v[pos];
  }
  ans = min(ans, llabs(a - b));
}
cout << ans << endl;
```

Two Pointers & Sliding Window Patterns `two_pointers` | O(N)

Opposite direction (2-Sum), same direction (Sliding Window), and merging 2 lists.

```
// 1. OPPOSITE DIRECTION (Target Sum on Sorted Array: O
(N))
int l = 0, r = n - 1;
while (l < r) {
    int sum = a[l] + a[r];
    if (sum == target) { /* found pair (l, r) */ break;
}
    else if (sum < target) l++;
    else r--;
}

// 2. SAME DIRECTION (sum <= K, a[i] >= 0 and K >= 0)
int left = 0, max_len = 0;
long long cur_sum = 0;
for (int right = 0; right < n; ++right) {
    cur_sum += a[right];
    while (cur_sum > K && left <= right) cur_sum -= a[le
ft++];
    max_len = max(max_len, right - left + 1);
}

// 3. MERGING TWO SORTED LISTS (Original twoPointer snip
pet: O(N + M))
vector<int> a(n), b(m), c;
for (int &x : a) cin >> x;
for (int &x : b) cin >> x;
int i = 0, j = 0;
while (i < n || j < m) {
    while ((i < n) && (j >= m || (a[i] <= b[j]))) { c.pu
sh_back(a[i]); i++; }
    while ((j < m) && (i >= n || b[j] <= a[i])) { c.push
_back(b[j]); j++; }
}
```

Binary Search on Answer (Integer & Real) `bs_ans` | O(log(Range) × Check)

Find optimal threshold mid via monotonic boolean condition check(mid).

```
// 1. Integer Range: Find minimum valid mid satisfying c
heck(mid)
long long lo = 1, hi = 1e18, best = -1;
while (lo <= hi) {
    long long mid = lo + (hi - lo) / 2;
    if (check(mid)) {
        best = mid;
        hi = mid - 1; // Try to find smaller valid thres
hold
    } else {
        lo = mid + 1; // Must go larger
    }
}

// 2. Real range: double stops at machine precision, not
10^-30.
double l = 0.0, r = 1e9;
for (int iter = 0; iter < 100; ++iter) {
    double mid = (l + r) / 2.0;
    if (mid == l || mid == r) break;
    if (check_double(mid)) r = mid; // or l = mid depend
ing on monotonicity
    else l = mid;
}
```

0-1 Knapsack & Unbounded Knapsack `O(N * W)`

```
int knapsack_01(int W, const vector<int>& wt, const vect
or<int>& val, int n) {
  vector<int> dp(W + 1, 0);
  for (int i = 0; i < n; i++)
    for (int w = W; w >= wt[i]; w--) // BACKWARDS for 0-
1
      dp[w] = max(dp[w], dp[w - wt[i]] + val[i]);
  return dp[W];
}
// Unbounded: loop forward: for (int w = wt[i]; w <= W;
w++)
```

Longest Increasing Subsequence (LIS) `O(N log N)`

```
int lis(const vector<int>& a) {
  vector<int> tails;
  for (int x : a) {
    auto it = lower_bound(tails.begin(), tails.end(),
x);
    if (it == tails.end()) tails.push_back(x);
    else *it = x;
  }
  return tails.size();
}
```

Kadane's Algorithm (Max Subarray Sum) `O(N)`

Maximum contiguous sum and [best_l, best_r] range in single pass.

```
long long max_sum = -1e18, cur = 0;
int best_l = 0, best_r = 0, start = 0;
for (int i = 0; i < n; i++) {
  cur += a[i];
  if (cur > max_sum) { max_sum = cur; best_l = start; be
st_r = i; }
  if (cur < 0) { cur = 0; start = i + 1; }
} // Maximum subarray sum = max_sum on range [best_l, be
st_r]
```

Longest Common Subsequence (LCS) `O(N × M)`

LCS length and string reconstruction via backtracking.

```
string get_lcs(const string &s, const string &t) {
  int n = s.size(), m = t.size();
  vector<vector<int>> dp(n + 1, vector<int>(m + 1, 0));
  for (int i = 1; i <= n; i++)
    for (int j = 1; j <= m; j++)
      dp[i][j] = (s[i-1] == t[j-1]) ? 1 + dp[i-1][j-1] :
max(dp[i-1][j], dp[i][j-1]);
  string res = ""; int i = n, j = m;
  while (i > 0 && j > 0) {
    if (s[i-1] == t[j-1]) { res += s[i-1]; i--; j--; }
    else if (dp[i-1][j] > dp[i][j-1]) i--;
    else j--;
  }
  reverse(res.begin(), res.end()); return res; // Length
= dp[n][m]
}
```

Coin Change (Min Coins & Total Ways) `O(N × Target)`

Min coins to make amount X, and number of combinations modulo MOD.

```
// 1. Min coins to make target X:
vector<int> dp(X + 1, 1e9); dp[0] = 0;
for (int i = 1; i <= X; i++)
  for (int c : coins) if (i >= c) dp[i] = min(dp[i], dp
[i - c] + 1);
int min_coins = (dp[X] >= 1e9 ? -1 : dp[X]);

// 2. Number of ways (Combinations - coins loop outsid
e!):
vector<int> ways(X + 1, 0); ways[0] = 1;
for (int c : coins)
  for (int i = c; i <= X; i++) ways[i] = (ways[i] + ways
[i - c]) % MOD;
```

Edit Distance (Levenshtein) `O(N × M)`

Minimum operations (insert, delete, replace) to transform s1 into s2.

```
int edit_distance(const string &s1, const string &s2) {
  int n = s1.size(), m = s2.size();
  vector<vector<int>> dp(n + 1, vector<int>(m + 1));
  for (int i = 0; i <= n; i++) dp[i][0] = i;
  for (int j = 0; j <= m; j++) dp[0][j] = j;
  for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= m; j++) {
      if (s1[i-1] == s2[j-1]) dp[i][j] = dp[i-1][j-1];
      else dp[i][j] = 1 + min({dp[i-1][j], dp[i][j-1], d
p[i-1][j-1]});
    }
  }
  return dp[n][m];
}
```

10. ADVANCED TOOLS (MST, MATRIX & GEOMETRY)

Kruskal's Minimum Spanning Tree (MST) `kruskal` | `O(E log E)`

Self-contained Kruskal's MST algorithm with internal DisjointSet (Union by Size & Path Compression), Edge struct, and main().

```
#include <bits/stdc++.h>
using namespace std;

class DisjointSet {
  vector<int> parent, size;
public:
  DisjointSet(int n) {
    parent.resize(n + 1);
    size.resize(n + 1, 1);
    for (int i = 0; i <= n; i++) parent[i] = i;
  }
  int findUPar(int node) {
    if (node == parent[node]) return node;
    return parent[node] = findUPar(parent[node]);
  }
  void unionBySize(int u, int v) {
    int ulpU = findUPar(u), ulpV = findUPar(v);
    if (ulpU == ulpV) return;
    if (size[ulpU] < size[ulpV]) {
      parent[ulpU] = ulpV; size[ulpV] += size[ulpU];
    } else {
      parent[ulpV] = ulpU; size[ulpU] += size[ulpV];
    }
  }
  bool connected(int u, int v) { return findUPar(u) == f
indUPar(v); }
};

struct Edge {
  int u, v, w;
  bool operator<(const Edge& o) const { return w < o.w;
}
};

optional<long long> kruskal(int n, vector<Edge>& edges)
{
  sort(edges.begin(), edges.end());
  DisjointSet dsu(n);
  long long cost = 0; int count = 0;
  if (n <= 1) return 0;
  for (auto& e : edges) {
    if (!dsu.connected(e.u, e.v)) {
      dsu.unionBySize(e.u, e.v);
      cost += e.w;
      count++;
      if (count == n - 1) return cost;
    }
  }
  return nullopt; // disconnected; a valid MST may have
cost -1
}

int main() {
  ios::sync_with_stdio(false);
  cin.tie(nullptr);
  int n, m; cin >> n >> m;
  vector<Edge> edges(m);
  for (int i = 0; i < m; i++) cin >> edges[i].u >> edges
[i].v >> edges[i].w;
  auto ans = kruskal(n, edges);
  if (!ans) cout << "MST does not exist\n";
  else cout << "Minimum cost = " << *ans << '\n';
  return 0;
}
```

Matrix Exponentiation (O(K^3 log N))

```
template<typename T = long long>
struct Matrix {
  int n; vector<vector<T>> a;
  Matrix(int n) : n(n), a(n, vector<T>(n, 0)) {}
  static Matrix eye(int n) {
    Matrix res(n); for (int i = 0; i < n; i++) res.a[i]
[i] = 1; return res;
  }
  Matrix operator*(const Matrix& o) const {
    Matrix res(n);
    for (int i = 0; i < n; i++)
      for (int k = 0; k < n; k++)
        for (int j = 0; j < n; j++)
          res.a[i][j] = (res.a[i][j] + a[i][k] * o.a[k]
[j]) % 1000000007;
    return res;
  }
  static Matrix power(Matrix m, long long p) {
    Matrix res = Matrix::eye(m.n);
    while (p) {
      if (p & 1) res = res * m;
      m = m * m; p >>= 1;
    }
    return res;
  }
};
```

Shoelace Formula (Polygon Area) `O(N)`

```
struct Point { double x, y; };
double polygon_area(const vector<Point>& p) {
  double area = 0.0; int n = p.size();
  for (int i = 0; i < n; ++i) {
    int j = (i + 1) % n;
    area += p[i].x * p[j].y - p[j].x * p[i].y;
  }
  return fabs(area) / 2.0;
}
```

2D Point, Vectors & Products `geom2d`

Vector operations, dot/cross products, CCW orientation, distance, angle.

```
struct Pt {
  double x, y;
  Pt(double x=0, double y=0) : x(x), y(y) {}
  Pt operator+(Pt p) const { return {x + p.x, y + p.y};
}
  Pt operator-(Pt p) const { return {x - p.x, y - p.y};
}
  Pt operator*(double d) const { return {x * d, y * d};
}
  Pt operator/(double d) const { return {x / d, y / d};
}
  double dot(Pt p) const { return x * p.x + y * p.y; }
// |a||b|cos(θ)
  double cross(Pt p) const { return x * p.y - y * p.x; }
// |a||b|sin(θ)
  double dist2() const { return x * x + y * y; }
  double dist() const { return sqrt(dist2()); }
  double angle() const { return atan2(y, x); }
};
// Orientation: > 0 = CCW (left turn), < 0 = CW (right t
urn), == 0 = Collinear
double ccw(Pt a, Pt b, Pt c) { return (b - a).cross(c -
a); }
```

Lines, Segments & Triangles Equations `line_tri`

Point-line distance, line intersection (Cramer), Heron, inradius, circumradius.

```
// 1. Distance from point P to line AB:
double dist_to_line(Pt p, Pt a, Pt b) {
  return fabs((b - a).cross(p - a)) / (b - a).dist();
}
// 2. Line Intersection (A1*x + B1*y = C1 and A2*x + B2*
y = C2 via Cramer):
bool line_inter(double a1, double b1, double c1, double
a2, double b2, double c2, Pt &out) {
  double d = a1 * b2 - a2 * b1;
  if (fabs(d) < 1e-9) return false; // Parallel or ident
ical
  out.x = (c1 * b2 - c2 * b1) / d; out.y = (a1 * c2 - a2
* c1) / d;
  return true;
}
// 3. Triangle Formulas: sides a, b, c
// Heron's Area: s = (a + b + c) / 2
double heron(double a, double b, double c) {
  double s = (a + b + c) / 2.0;
  return sqrt(max(0.0, s * (s - a) * (s - b) * (s -
c)));
}
// Inradius: r = Area / s | Circumradius: R = (a * b *
c) / (4 * Area)
double inradius(double a, double b, double c) { return h
eron(a, b, c) / ((a + b + c) / 2.0); }
double circumradius(double a, double b, double c) { retu
rn (a * b * c) / (4.0 * heron(a, b, c)); }
```

Pick's Theorem, Centroid & 3D Solids `geom_extra`

Pick's Theorem, polygon centroid, circle, and 3D solid geometry equations.

```
// 1. Pick's Theorem: Area = I + B/2 - 1  =>  I = Area -
B/2 + 1
// B = boundary lattice points = sum(gcd(|x2-x1|, |y2-y1
|)) for all polygon edges
// I = strictly interior lattice points

// 2. Polygon Centroid (Center of Mass):
Pt polygon_centroid(const vector<Pt>& p) {
  Pt c(0, 0); double A = 0; int n = p.size();
  for (int i = 0; i < n; i++) {
    int j = (i + 1) % n; double cross = p[i].cross(p
[j]);
    A += cross; c = c + (p[i] + p[j]) * cross;
  }
  return c / (3.0 * A);
}

// 3. Circles & 3D Solid Geometry Equations:
// Circle:   Area = π * r^2, Circumference = 2 * π * r
// Sphere:   Volume = (4/3) * π * r^3, Surface Area = 4
* π * r^2
// Cylinder: Volume = π * r^2 * h, Surface Area = 2 * π
* r * (r + h)
// Cone:     Volume = (1/3) * π * r^2 * h, Slant height
s = sqrt(r^2 + h^2)
```

Math Series, Logarithms & Algebra `math_eq`

Closed-form summations, geometric progressions, logarithm laws, digit counts.

```
// 1. Summation Formulas:
// Sum of 1..N:           N * (N + 1) / 2
// Sum of i^2 for i=1..N: N * (N + 1) * (2N + 1) / 6
// Sum of i^3 for i=1..N: (N * (N + 1) / 2)^2
// Geometric Progression: a + a*r + ... + a*r^(n-1) = a
* (r^n - 1) / (r - 1)

// 2. Logarithm & Exponents:
// log_b(a) = ln(a) / ln(b) = log10(a) / log10(b)
// log(a * b) = log(a) + log(b), log(a / b) = log(a) - l
og(b), log(a^k) = k * log(a)

// 3. Number of Digits of N in Base B: floor(log_B(N)) +
1
// 4. Stirling's Approximation: n! ≈ sqrt(2 * π * n) *
(n / e)^n
```

11. GRAPH ALGORITHMS

Strongly connected components `KACT graph pp. 12–13`

O(V+E); IDs are source-first. Use for condensation DAGs.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct SCC {
  int n;
  std::vector<std::vector<int>> g, rg;
  explicit SCC(int n) : n(n), g(n), rg(n) {}
  void add_edge(int u, int v) {
    g[u].push_back(v);
    rg[v].push_back(u);
  }

  // Kosaraju: component IDs are in topological order (s
ource first).
  std::vector<int> components() const {
    std::vector<char> seen(n);
    std::vector<int> order, comp(n, -1);
    for (int root = 0; root < n; ++root)
      if (!seen[root]) {
        std::vector<std::pair<int, int>> st{{root, 0}};
        seen[root] = 1;
        while (!st.empty()) {
          auto &[u, i] = st.back();
          if (i == (int)g[u].size()) {
            order.push_back(u);
            st.pop_back();
          } else {
            int v = g[u][i++];
            if (!seen[v])
              seen[v] = 1, st.push_back({v, 0});
          }
        }
      }
    int id = 0;
    for (int i = n - 1; i >= 0; --i) {
      int root = order[i];
      if (comp[root] != -1)
        continue;
      std::vector<int> st{root};
      comp[root] = id;
      while (!st.empty()) {
        int u = st.back();
        st.pop_back();
        for (int v : rg[u])
          if (comp[v] == -1)
            comp[v] = id, st.push_back(v);
      }
      ++id;
    }
    return comp;
  }
};
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

2-SAT

KACT graph p. 13

O(V+E). Add each clause (x=value OR y=value); depends on SCC.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
// Depends on SCC above. Literal (variable x, truth valu
e val) is 2*x +
// val.
struct TwoSAT {
  int n;
  SCC scc;
  explicit TwoSAT(int n) : n(n), scc(2 * n) {}
  int lit(int x, bool val) const { return 2 * x + int(va
l); }
  void implication(int a, int b) { scc.add_edge(a, b); }
  void add_or(int x, bool xv, int y, bool yv) {
    int a = lit(x, xv), b = lit(y, yv);
    implication(a ^ 1, b);
    implication(b ^ 1, a);
  }
  std::optional<std::vector<bool>> solve() const {
    auto c = scc.components();
    std::vector<bool> ans(n);
    for (int x = 0; x < n; ++x) {
      if (c[2 * x] == c[2 * x + 1])
        return std::nullopt;
      ans[x] = c[2 * x + 1] > c[2 * x];
    }
    return ans;
  }
};
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Bridges in an undirected multigraph

KACT graph p. 12

O(V+E). Give every edge a unique ID; recursive DFS has a stack-depth limit.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
// Undirected multigraph: edge IDs prevent a parallel ed
ge becoming a
// false bridge.
struct Bridges {
  int n, timer = 0;
  std::vector<std::vector<std::pair<int, int>>> g;
  std::vector<int> tin, low, bridge_ids;
  explicit Bridges(int n) : n(n), g(n), tin(n, -1), low
(n) {}
  void add_edge(int u, int v, int id) {
    g[u].push_back({v, id});
    g[v].push_back({u, id});
  }
  void dfs(int u, int parent_edge = -1) {
    tin[u] = low[u] = timer++;
    for (auto [v, id] : g[u]) {
      if (id == parent_edge)
        continue;
      if (tin[v] != -1)
        low[u] = std::min(low[u], tin[v]);
      else {
        dfs(v, id);
        low[u] = std::min(low[u], low[v]);
        if (low[v] > tin[u])
          bridge_ids.push_back(id);
      }
    }
  }
  std::vector<int> find() {
    timer = 0;
    std::fill(tin.begin(), tin.end(), -1);
    std::fill(low.begin(), low.end(), 0);
    bridge_ids.clear();
    for (int u = 0; u < n; ++u)
      if (tin[u] == -1)
        dfs(u);
    return bridge_ids;
  }
}; // Recursive DFS: use an iterative version if depth m
ay exceed stack
   // capacity.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Dinic maximum flow

KACT graph p. 10

Directed capacities, 0-indexed vertices; add both directions explicitly if needed.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct Dinic {
  struct Edge {
    int to, rev;
    long long cap;
  };
  int n;
  std::vector<std::vector<Edge>> g;
  std::vector<int> level, ptr;
  explicit Dinic(int n) : n(n), g(n), level(n), ptr(n)
{}
  void add_edge(int u, int v, long long cap) { // direct
ed, cap >= 0
    if (u == v)
      return;
    int a = (int)g[u].size(), b = (int)g[v].size();
    g[u].push_back({v, b, cap});
    g[v].push_back({u, a, 0});
  }
  bool bfs(int s, int t) {
    std::fill(level.begin(), level.end(), -1);
    std::queue<int> q;
    level[s] = 0;
    q.push(s);
    while (!q.empty()) {
      int u = q.front();
      q.pop();
      for (auto &e : g[u])
        if (e.cap > 0 && level[e.to] < 0)
          level[e.to] = level[u] + 1, q.push(e.to);
    }
    return level[t] >= 0;
  }
  long long dfs(int u, int t, long long f) {
    if (u == t || f == 0)
      return f;
    for (int &i = ptr[u]; i < (int)g[u].size(); ++i) {
      Edge &e = g[u][i];
      if (e.cap == 0 || level[e.to] != level[u] + 1)
        continue;
      long long sent = dfs(e.to, t, std::min(f, e.cap));
      if (sent) {
        e.cap -= sent;
        g[e.to][e.rev].cap += sent;
        return sent;
      }
    }
    return 0;
  }
  long long max_flow(int s, int t) {
    if (s == t)
      return 0;
    long long ans = 0;
    while (bfs(s, t)) {
      std::fill(ptr.begin(), ptr.end(), 0);
      while (long long f = dfs(s, t, LLONG_MAX))
        ans += f;
    }
    return ans;
  }
}; // O(V^2 E) general bound. DFS is recursive.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Hopcroft–Karp matching

KACT graph p. 11

O(E√V). Left and right vertices are numbered separately from zero.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct HopcroftKarp {
  int nl, nr;
  std::vector<std::vector<int>> g;
  std::vector<int> left, right, dist;
  int shortest;
  HopcroftKarp(int nl, int nr)
      : nl(nl), nr(nr), g(nl), left(nl, -1), right(nr, -
1), dist(nl) {}
  void add_edge(int u, int v) {
    g[u].push_back(v);
  } // u in left, v in right
  bool bfs() {
    std::queue<int> q;
    shortest = INT_MAX;
    std::fill(dist.begin(), dist.end(), -1);
    for (int u = 0; u < nl; ++u)
      if (left[u] == -1)
        dist[u] = 0, q.push(u);
    while (!q.empty()) {
      int u = q.front();
      q.pop();
      if (dist[u] >= shortest)
        continue;
      for (int v : g[u]) {
        int w = right[v];
        if (w == -1)
          shortest = dist[u] + 1;
        else if (dist[w] == -1)
          dist[w] = dist[u] + 1, q.push(w);
      }
    }
    return shortest != INT_MAX;
  }
  bool dfs(int u) {
    for (int v : g[u]) {
      int w = right[v];
      if ((w == -1 && dist[u] + 1 == shortest) ||
          (w != -1 && dist[w] == dist[u] + 1 && dfs(w)))
{
        left[u] = v;
        right[v] = u;
        return true;
      }
    }
    dist[u] = -1;
    return false;
  }
  int max_matching() {
    int ans = 0;
    for (int v : left) ans += (v != -1);
    while (bfs())
      for (int u = 0; u < nl; ++u)
        if (left[u] == -1 && dfs(u))
          ++ans;
    return ans;
  }
}; // O(E sqrt(V)); recursion may be deep.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Undirected Euler walk

KACT graph p. 13

O(V+E). Supports parallel edges and loops; returns no path if degree/connectivity conditions fail.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
// Euler path/circuit of an undirected multigraph; retur
ns vertices or
// nullopt.
struct EulerWalk {
  int n, edges = 0;
  std::vector<std::vector<std::pair<int, int>>> g;
  explicit EulerWalk(int n) : n(n), g(n) {}
  void add_edge(int u, int v) {
    g[u].push_back({v, edges});
    g[v].push_back({u, edges});
    ++edges;
  }
  std::optional<std::vector<int>> walk() const {
    if (!n)
      return std::vector<int>{};
    int start = -1, odd = 0;
    for (int u = 0; u < n; ++u) {
      if (g[u].size() % 2)
        ++odd, start = u;
      else if (start == -1 && !g[u].empty())
        start = u;
    }
    if (odd != 0 && odd != 2)
      return std::nullopt;
    if (start == -1)
      return std::vector<int>{0};
    std::vector<int> ptr(n), stack{start}, path;
    std::vector<char> used(edges);
    while (!stack.empty()) {
      int u = stack.back();
      while (ptr[u] < (int)g[u].size() && used[g[u][ptr
[u]].second])
        ++ptr[u];
      if (ptr[u] == (int)g[u].size())
        path.push_back(u), stack.pop_back();
      else {
        auto [v, id] = g[u][ptr[u]++];
        if (!used[id])
          used[id] = 1, stack.push_back(v);
      }
    }
    if ((int)path.size() != edges + 1)
      return std::nullopt; // disconnected edges
    std::reverse(path.begin(), path.end());
    return path;
  }
}; // O(V+E). Isolated vertices are irrelevant when edge
s exist.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

12. OFFLINE QUERIES AND ASSIGNMENT

Mo's algorithm

Notebook p. 2; linked GitHub Mo skeleton

Offline inclusive [l,r] queries. Supply inverse add/remove callbacks and a current-answer callback.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct MoQuery {
  int l, r, id;
}; // inclusive [l,r], IDs 0..Q-1
template <class Ans, class Add, class Remove, class Get>
std::vector<Ans> mo_process(int n, std::vector<MoQuery>
qs, Add add,
                            Remove remove, Get get) {
  int block = std::max(1, (int)std::sqrt(std::max(1,
n)));
  std::sort(
      qs.begin(), qs.end(), [&](const MoQuery &a, const
MoQuery &b) {
        int x = a.l / block, y = b.l / block;
        if (x != y)
          return x < y;
        return (x & 1)
                   ? a.r > b.r
                   : a.r < b.r; // snake order limits po
inter movement
      });
  std::vector<Ans> result(qs.size());
  int l = 0, r = -1; // current window is [l,r], initial
ly empty
  for (auto q : qs) {
    while (l > q.l)
      add(--l);
    while (r < q.r)
      add(++r);
    while (l < q.l)
      remove(l++);
    while (r > q.r)
      remove(r--);
    result[q.id] = get();
  }
  return result;
} // Add/remove must be inverse; static array, offline q
ueries.
// Adapted from Mohammadreza-mz's Mo Algorithm.cpp and t
he supplied
// notebook.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Hungarian minimum-cost assignment

Notebook pp. 14–15

O(n²m), n≤m; negative costs work. Returns total cost and a 0-indexed column for every row.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
// Minimum-cost assignment; n rows, m columns, n<=m. Eve
ry row gets one
// distinct column.
struct Assignment {
  long long cost;
  std::vector<int> column_of_row;
};
inline Assignment
hungarian(const std::vector<std::vector<long long>> &a)
{
  int n = (int)a.size();
  if (!n)
    return {0, {}};
  int m = (int)a[0].size();
  assert(n <= m && m > 0);
  for (auto &row : a)
    assert((int)row.size() == m);
  const long long INF = LLONG_MAX / 4;
  std::vector<long long> u(n + 1), v(m + 1);
  std::vector<int> p(m + 1), way(m + 1);
  for (int i = 1; i <= n; ++i) {
    p[0] = i;
    int j0 = 0;
    std::vector<long long> minv(m + 1, INF);
    std::vector<char> used(m + 1);
    do {
      used[j0] = 1;
      int i0 = p[j0], j1 = 0;
      long long delta = INF;
      for (int j = 1; j <= m; ++j)
        if (!used[j]) {
          long long cur = a[i0 - 1][j - 1] - u[i0] - v
[j];
          if (cur < minv[j])
            minv[j] = cur, way[j] = j0;
          if (minv[j] < delta)
            delta = minv[j], j1 = j;
        }
      for (int j = 0; j <= m; ++j)
        if (used[j])
          u[p[j]] += delta, v[j] -= delta;
        else
          minv[j] -= delta;
      j0 = j1;
    } while (p[j0] != 0);
    do {
      int j1 = way[j0];
      p[j0] = p[j1];
      j0 = j1;
    } while (j0);
  }
  std::vector<int> col(n, -1);
  for (int j = 1; j <= m; ++j)
    if (p[j])
      col[p[j] - 1] = j - 1;
  return {-v[0], col};
} // O(n^2 m), handles negative costs. Use long long cos
ts with safe
  // total bounds.
// Adapted from the supplied notebook, pp. 14-15.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

13. STRING ALGORITHMS

Z function

KACT strings p. 22

O(N). z[i] is the longest prefix matching the substring beginning at i; z[0]=0.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
inline std::vector<int> z_function(const std::string &s)
{
  int n = (int)s.size();
  std::vector<int> z(n);
  for (int i = 1, l = 0, r = 0; i < n; ++i) {
    if (i < r)
      z[i] = std::min(r - i, z[i - l]);
    while (i + z[i] < n && s[z[i]] == s[i + z[i]])
      ++z[i];
    if (i + z[i] > r)
      l = i, r = i + z[i];
  }
  return z; // z[0]=0; Z box is [l,r).
}
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Manacher palindrome radii

KACT strings p. 22

O(N). Separate odd and even radii; formulas for actual lengths are in the code.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct PalRadii {
  std::vector<int> odd, even;
};
inline PalRadii manacher(const std::string &s) {
  int n = (int)s.size();
  PalRadii p{{}, {}};
  p.odd.resize(n);
  p.even.resize(n);
  for (int i = 0, l = 0, r = -1; i < n; ++i) {
    int k = i > r ? 1 : std::min(p.odd[l + r - i], r - i
+ 1);
    while (i - k >= 0 && i + k < n && s[i - k] == s[i +
k])
      ++k;
    p.odd[i] = k;
    if (i + k - 1 > r)
      l = i - k + 1, r = i + k - 1;
  }
  for (int i = 0, l = 0, r = -1; i < n; ++i) {
    int k = i > r ? 0 : std::min(p.even[l + r - i + 1],
r - i + 1);
    while (i - k - 1 >= 0 && i + k < n && s[i - k - 1] =
= s[i + k])
      ++k;
    p.even[i] = k;
    if (i + k - 1 > r)
      l = i - k, r = i + k - 1;
  }
  return p; // odd palindrome length = 2*odd[i]-1; even
= 2*even[i].
}
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Aho–Corasick multiple-pattern matching

KACT strings p. 23, generalized

Lowercase a–z patterns; count overlaps and duplicates. Build after all insertions.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct AhoCorasick {
  struct Node {
    std::array<int, 26> next{};
    int link = 0, out = 0;
  };
  std::vector<Node> t{1};
  bool built = false;
  void
  add(const std::string &s) { // nonempty lowercase a-z
patterns only
    assert(!built && !s.empty());
    int u = 0;
    for (char ch : s) {
      assert('a' <= ch && ch <= 'z');
      int c = ch - 'a';
      if (!t[u].next[c]) {
        t[u].next[c] = (int)t.size();
        t.emplace_back();
      }
      u = t[u].next[c];
    }
    ++t[u].out; // duplicates count separately
  }
  void build() {
    assert(!built);
    std::queue<int> q;
    for (int c = 0; c < 26; ++c)
      if (t[0].next[c])
        q.push(t[0].next[c]);
    while (!q.empty()) {
      int u = q.front();
      q.pop();
      t[u].out += t[t[u].link].out;
      for (int c = 0; c < 26; ++c) {
        int v = t[u].next[c];
        if (v)
          t[v].link = t[t[u].link].next[c], q.push(v);
        else
          t[u].next[c] = t[t[u].link].next[c];
      }
    }
    built = true;
  }
  long long count_matches(const std::string &text) const
{
    assert(built);
    long long ans = 0;
    int u = 0;
    for (char ch : text) {
      if (ch < 'a' || ch > 'z') {
        u = 0;
        continue;
      }
      u = t[u].next[ch - 'a'];
      ans += t[u].out;
    }
    return ans;
  }
}; // O(total pattern length * alphabet + text length).
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Suffix array and LCP

KACT strings p. 22, simplified

O(N log²N). Uses sorting for clarity; use a radix version only after benchmarking.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct SuffixArray {
  std::vector<int> sa,
      lcp; // sa has n entries, lcp[i]=LCP(sa[i],sa[i-
1]), lcp[0]=0
  explicit SuffixArray(const std::string &s) {
    int n = (int)s.size();
    sa.resize(n);
    lcp.assign(n, 0);
    if (!n)
      return;
    std::vector<int> rank(n), tmp(n), pos(n);
    for (int i = 0; i < n; ++i)
      sa[i] = i, rank[i] = (unsigned char)s[i];
    for (int k = 1; k < n; k *= 2) {
      auto key = [&](int i) {
        return std::pair<int, int>{rank[i],
                                   i + k < n ? rank[i +
k] : -1};
      };
      std::sort(sa.begin(), sa.end(),
                [&](int a, int b) { return key(a) < key
(b); });
      tmp[sa[0]] = 0;
      for (int i = 1; i < n; ++i)
        tmp[sa[i]] = tmp[sa[i - 1]] + (key(sa[i - 1]) !=
key(sa[i]));
      rank.swap(tmp);
      if (rank[sa.back()] == n - 1)
        break;
    }
    for (int i = 0; i < n; ++i)
      pos[sa[i]] = i;
    for (int i = 0, k = 0; i < n; ++i) {
      int r = pos[i];
      if (!r)
        continue;
      int j = sa[r - 1];
      while (i + k < n && j + k < n && s[i + k] == s[j +
k])
        ++k;
      lcp[r] = k;
      if (k)
        --k;
    }
  }
}; // O(n log^2 n) with std::sort; simpler to audit than
radix version.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Static double rolling hash

KACT strings p. 23, range convention standardized

O(N) build and O(1) substring hash. Uses half-open [l,r) ranges; collisions remain possible.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
// Static double rolling hash of half-open string ranges
[l,r).
struct DoubleHash {
  static constexpr long long P1 = 1000000007LL, P2 = 100
0000009LL,
                              B = 911382323LL;
  std::vector<long long> h1, h2, p1, p2;
  explicit DoubleHash(const std::string &s) {
    int n = (int)s.size();
    h1.resize(n + 1);
    h2.resize(n + 1);
    p1.resize(n + 1);
    p2.resize(n + 1);
    p1[0] = p2[0] = 1;
    for (int i = 0; i < n; ++i) {
      int ch = (unsigned char)s[i] + 1;
      h1[i + 1] = (h1[i] * B + ch) % P1;
      h2[i + 1] = (h2[i] * B + ch) % P2;
      p1[i + 1] = p1[i] * B % P1;
      p2[i + 1] = p2[i] * B % P2;
    }
  }
  std::pair<long long, long long> get(int l, int r) cons
t {
    assert(0 <= l && l <= r && r < (int)h1.size());
    return {(h1[r] - h1[l] * p1[r - l] % P1 + P1) % P1,
            (h2[r] - h2[l] * p2[r - l] % P2 + P2) % P2};
  }
}; // O(N) build, O(1) substring hash; collision probabi
lity is low, not
   // zero.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

14. NUMBER THEORY

Extended GCD and general CRT

KACT number theory p. 9, checked variant

Handles non-coprime moduli; returns no solution on incompatibility or 64-bit lcm overflow.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
inline long long extended_gcd(long long a, long long b,
long long &x,
                              long long &y) {
  if (!b) {
    x = 1;
    y = 0;
    return a;
  }
  long long x1, y1, g = extended_gcd(b, a % b, x1, y1);
  x = y1;
  y = x1 - (a / b) * y1;
  return g;
}
// Return {smallest nonnegative solution, lcm}; nullopt
if incompatible
// or overflow.
inline std::optional<std::pair<long long, long long>>
crt(long long a, long long m, long long b, long long n)
{
  assert(m > 0 && n > 0);
  a %= m;
  if (a < 0)
    a += m;
  b %= n;
  if (b < 0)
    b += n;
  long long x, y, g = extended_gcd(m, n, x, y);
  __int128 diff = (__int128)b - a;
  if (diff % g)
    return std::nullopt;
  __int128 mod = (__int128)m / g * n;
  if (mod > LLONG_MAX)
    return std::nullopt;
  __int128 k = (diff / g) * x;
  long long ng = n / g;
  k = (k % ng + ng) % ng;
  long long rem = (long long)((a + k * m) % mod);
  return std::pair<long long, long long>{rem, (long lon
g)mod};
}
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Linear Euler totient sieve

KACT number theory p. 9

O(N) precomputation; phi[1]=1. Useful for coprime counting.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
inline std::vector<int> phi_sieve(int n) { // phi[1]=1,
phi[0]=0
  assert(n >= 0);
  std::vector<int> phi(n + 1), primes;
  std::vector<char> composite(n + 1);
  if (n >= 1)
    phi[1] = 1;
  for (int i = 2; i <= n; ++i) {
    if (!composite[i])
      primes.push_back(i), phi[i] = i - 1;
    for (int p : primes) {
      if (1LL * i * p > n)
        break;
      composite[i * p] = 1;
      if (i % p == 0) {
        phi[i * p] = phi[i] * p;
        break;
      }
      phi[i * p] = phi[i] * (p - 1);
    }
  }
  return phi;
} // O(n) time and memory.
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

64-bit Miller–Rabin

KACT number theory p. 9, checked variant

Deterministic uint64 primality with __int128 modular multiplication; GNU C++17.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
using u64 = uint64_t;
using u128 = __uint128_t;
inline u64 mul_mod(u64 a, u64 b, u64 mod) { return (u12
8)a * b % mod; }
inline u64 pow_mod(u64 a, u64 e, u64 mod) {
  assert(mod > 0);
  u64 r = 1 % mod;
  for (; e; e >>= 1, a = mul_mod(a, a, mod))
    if (e & 1)
      r = mul_mod(r, a, mod);
  return r;
}
inline bool is_prime(u64 n) { // deterministic for every
uint64_t
  if (n < 2)
    return false;
  for (u64 p : {2ULL, 3ULL, 5ULL, 7ULL, 11ULL, 13ULL, 17
ULL, 19ULL,
                23ULL, 29ULL, 31ULL, 37ULL}) {
    if (n % p == 0)
      return n == p;
  }
  u64 d = n - 1, s = 0;
  while (!(d & 1))
    d >>= 1, ++s;
  for (u64 a : {2ULL, 325ULL, 9375ULL, 28178ULL, 450775U
LL, 9780504ULL,
                1795265022ULL}) {
    if (a % n == 0)
      continue;
    u64 x = pow_mod(a % n, d, n);
    if (x == 1 || x == n - 1)
      continue;
    bool passed = false;
    for (u64 r = 1; r < s; ++r) {
      x = mul_mod(x, x, n);
      if (x == n - 1) {
        passed = true;
        break;
      }
    }
    if (!passed)
      return false;
  }
  return true;
}
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

15. GEOMETRY ALGORITHMS

Exact integer geometry

KACT geometry pp. 15–17, unified

__int128 orientation; segment intersection, convex hull, twice signed area, point in polygon.

```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct IPoint {
  long long x, y;
  bool operator<(const IPoint &b) const {
    return std::tie(x, y) < std::tie(b.x, b.y);
  }
  bool operator==(const IPoint &b) const {
    return x == b.x && y == b.y;
  }
};
inline __int128 cross(IPoint a, IPoint b, IPoint c) {
  // Requires the determinant to fit signed __int128.
  return ((__int128)b.x - a.x) * ((__int128)c.y - a.y) -
         ((__int128)b.y - a.y) * ((__int128)c.x - a.x);
}
inline int sign(__int128 x) { return (x > 0) - (x < 0);
}
inline bool on_segment(IPoint a, IPoint b, IPoint p) {
  return cross(a, b, p) == 0 && std::min(a.x, b.x) <= p.
x &&
         p.x <= std::max(a.x, b.x) && std::min(a.y, b.y)
<= p.y &&
         p.y <= std::max(a.y, b.y);
}
inline bool segments_intersect(IPoint a, IPoint b, IPoin
t c, IPoint d) {
  int x = sign(cross(a, b, c)), y = sign(cross(a, b,
d));
  int z = sign(cross(c, d, a)), w = sign(cross(c, d,
b));
  if (x == 0 && on_segment(a, b, c))
    return true;
  if (y == 0 && on_segment(a, b, d))
    return true;
  if (z == 0 && on_segment(c, d, a))
    return true;
  if (w == 0 && on_segment(c, d, b))
    return true;
  return x != y && z != w;
}
inline std::vector<IPoint> convex_hull(std::vector<IPoin
t> p) {
  std::sort(p.begin(), p.end());
  p.erase(std::unique(p.begin(), p.end()), p.end());
  if (p.size() < 2)
    return p;
  std::vector<IPoint> h;
  for (auto v : p) {
    while (h.size() >= 2 && cross(h[h.size() - 2], h.bac
k(), v) <= 0)
      h.pop_back();
    h.push_back(v);
  }
  size_t lower = h.size();
  for (int i = (int)p.size() - 2; i >= 0; --i) {
    auto v = p[i];
    while (h.size() > lower && cross(h[h.size() - 2], h.
back(), v) <= 0)
      h.pop_back();
    h.push_back(v);
  }
  h.pop_back();
  return h; // CCW, starts at lexicographic min; colline
ar interior
            // removed.
}
inline __int128
area2(const std::vector<IPoint> &p) { // signed twice ar
ea
  __int128 a = 0;
  int n = (int)p.size();
  for (int i = 0; i < n; ++i)
    a += (__int128)p[i].x * p[(i + 1) % n].y -
         (__int128)p[i].y * p[(i + 1) % n].x;
  return a;
}
inline int point_in_polygon(const std::vector<IPoint> &
p, IPoint q) {
  bool inside = false;
  int n = (int)p.size();
  for (int i = 0, j = n - 1; i < n; j = i++) {
    IPoint a = p[j], b = p[i];
    if (on_segment(a, b, q))
      return 0; // boundary
    if ((a.y > q.y) != (b.y > q.y)) {
      // Compare crossing x without floating point; orie
ntation flips
      // with edge direction.
      __int128 c = cross(a, b, q);
      if ((c > 0) == (b.y > a.y))
        inside = !inside;
    }
  }
  return inside ? 1 : -1; // inside=1, boundary=0, outsi
de=-1
}
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

Circle–line intersection

KACT geometry p. 16

Infinite line AB; returns 0, 1, or 2 points. Rejects degenerate lines.

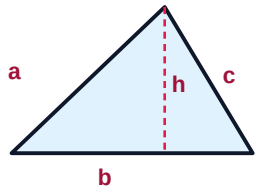
```
#ifdef int
#pragma push_macro("int")
#undef int
#define CPREF_POP_INT
#endif
struct FPoint {
  long double x, y;
};
inline FPoint operator+(FPoint a, FPoint b) {
  return {a.x + b.x, a.y + b.y};
}
inline FPoint operator-(FPoint a, FPoint b) {
  return {a.x - b.x, a.y - b.y};
}
inline FPoint operator*(FPoint a, long double k) {
  return {a.x * k, a.y * k};
}
inline long double dot(FPoint a, FPoint b) {
  return a.x * b.x + a.y * b.y;
}
// Intersection of infinite line AB and circle (center
O, radius r).
inline std::vector<FPoint> circle_line(FPoint o, long do
uble r,
                                       FPoint a, FPoint
b) {
  assert(r >= 0);
  FPoint d = b - a;
  long double len2 = dot(d, d);
  if (len2 == 0)
    return {}; // degenerate line
  long double t = dot(o - a, d) / len2;
  FPoint foot = a + d * t;
  long double dist2 = dot(foot - o, foot - o);
  long double h2 = r * r - dist2;
  // Relative tolerance on squared distances preserves v
ery small circles.
  long double tol = 64 * std::numeric_limits<long double
>::epsilon() *
                    std::max(r * r, dist2);
  if (h2 < -tol)
    return {};
  if (h2 <= tol)
    return {foot};
  long double step = std::sqrt(h2 / len2);
  return {foot - d * step, foot + d * step};
}
#ifdef CPREF_POP_INT
#pragma pop_macro("int")
#undef CPREF_POP_INT
#endif
```

16. ILLUSTRATED GEOMETRY FORMULAS

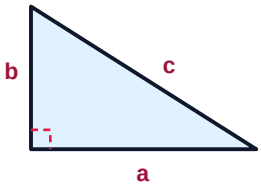
Symbols in each figure match the formula table. All angles are in radians unless degrees are stated.

2D shapes

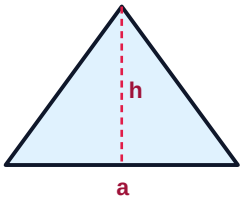
Triangle



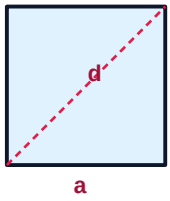
Right triangle



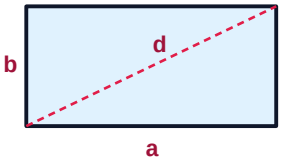
Equilateral triangle



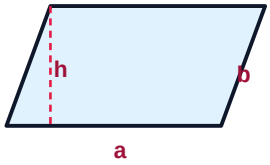
Square



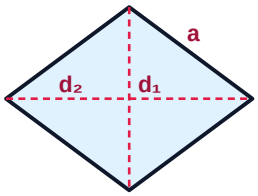
Rectangle



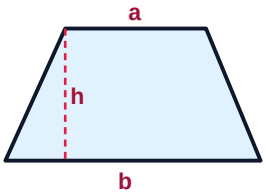
Parallelogram



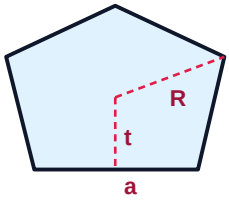
Rhombus



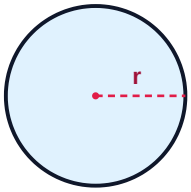
Trapezoid



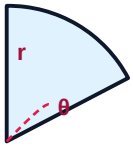
Regular n-gon



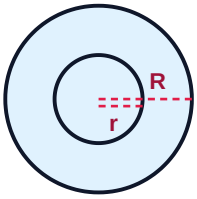
Circle



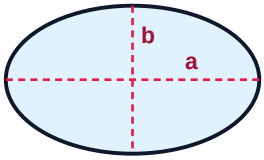
Sector



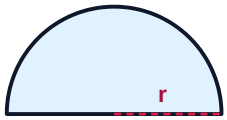
Annulus



Ellipse



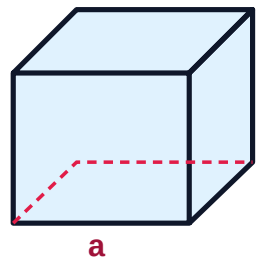
Semicircle



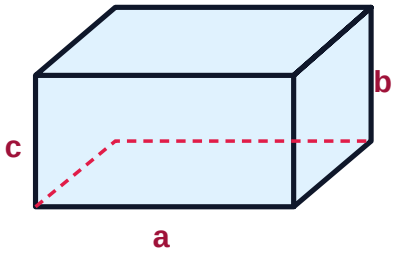
Shape / rule	Meaning	Formulas
Triangle	sides a,b,c; base b, height h	$A=bh/2=\sqrt{(s(s-a)(s-b)(s-c))}$, $s=(a+b+c)/2$; $P=a+b+c$; $r=A/s$; $R=abc/(4A)$.
Right triangle	legs a,b; hypotenuse c	$c^2=a^2+b^2$; $A=ab/2$; $R=c/2$; $r=(a+b-c)/2$.
Equilateral triangle	side a	$A=\sqrt{3}\cdot a^2/4$; $P=3a$; $h=\sqrt{3}\cdot a/2$; $r=\sqrt{3}\cdot a/6$; $R=\sqrt{3}\cdot a/3$.
Square	side a	$A=a^2$; $P=4a$; diagonal $d=a\sqrt{2}$; inradius $a/2$; circumradius $a/\sqrt{2}$.
Rectangle	sides a,b	$A=ab$; $P=2(a+b)$; diagonal $d=\sqrt{(a^2+b^2)}$.
Parallelogram	adjacent sides a,b; height h to a; angle θ	$A=ah=ab\sin\theta$; $P=2(a+b)$.
Rhombus	side a; height h; diagonals d_1,d_2	$A=d_1d_2/2=ah$; $P=4a$; diagonals bisect at right angles.
Trapezoid	parallel bases a,b; height h	$A=(a+b)h/2$; midline$=(a+b)/2$; P=sum of four sides.
Regular n-gon	n sides of length a; apothem t	$P=na$; $t=a/(2\tan(\pi/n))$; $A=Pt/2=na^2/(4\tan(\pi/n))$; circumradius $R=a/(2\sin(\pi/n))$.
Circle	radius r, diameter $d=2r$	$A=\pi r^2$; circumference $C=2\pi r$; chord length at central angle θ: $2r\sin(\theta/2)$.
Sector	radius r; angle θ in radians	$A=r^2\theta/2$; arc length$=r\theta$; perimeter$=2r+r\theta$.
Annulus	outer radius R; inner radius r	$A=\pi(R^2-r^2)$; total boundary length$=2\pi(R+r)$.
Ellipse	semiaxes $a\geq b$	$A=\pi ab$; focal distance $c=\sqrt{(a^2-b^2)}$; perimeter $\approx \pi[3(a+b)-\sqrt{((3a+b)(a+3b))}]$.
Semicircle	radius r	$A=\pi r^2/2$; perimeter including diameter$=\pi r+2r$.

3D solids

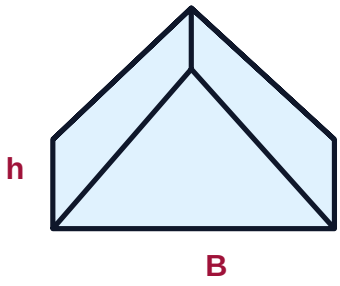
Cube



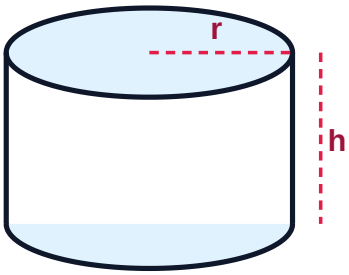
Cuboid



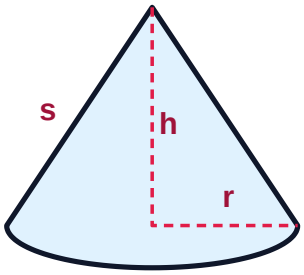
Prism



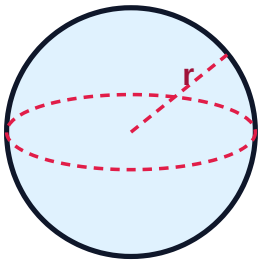
Cylinder



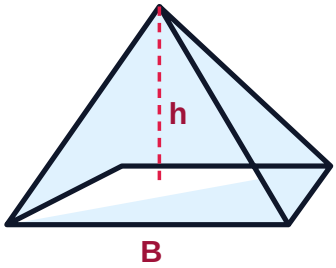
Cone



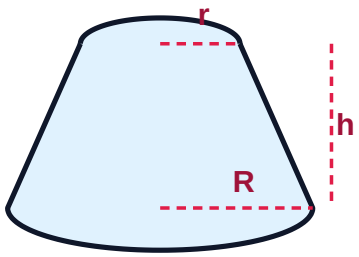
Sphere



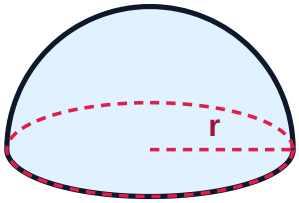
Pyramid



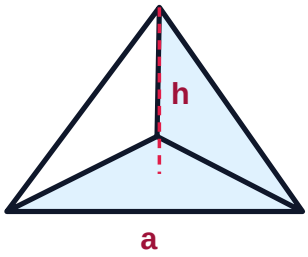
Frustum



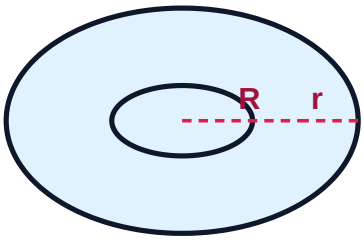
Hemisphere



Tetrahedron



Torus



Shape / rule	Meaning	Formulas
Cube	edge a	$V=a^3$; surface $S=6a^2$; space diagonal $=a\sqrt{3}$.
Cuboid	edges a,b,c	$V=abc$; $S=2(ab+bc+ca)$; space diagonal $=\sqrt{a^2+b^2+c^2}$.
Right prism	base area B; base perimeter P; height h	$V=Bh$; $S=2B+Ph$.
Cylinder	radius r; height h	$V=\pi r^2h$; curved area $=2\pi rh$; total $S=2\pi r(r+h)$.
Cone	radius r; height h; slant $s=\sqrt{r^2+h^2}$	$V=\pi r^2h/3$; curved area $=\pi rs$; total $S=\pi r(r+s)$.
Sphere	radius r	$V=4\pi r^3/3$; $S=4\pi r^2$.
Hemisphere	radius r	$V=2\pi r^3/3$; curved area $=2\pi r^2$; total $S=3\pi r^2$.
Right pyramid	base area B; perimeter P; height h; common face slant s	$V=Bh/3$; $S=B+Ps/2$ when every lateral face shares slant height s.
Cone frustum	radii $R\geq r$; height h; slant $s=\sqrt{h^2+(R-r)^2}$	$V=\pi h(R^2+Rr+r^2)/3$; $S=\pi(R^2+r^2)+\pi(R+r)s$.
Regular tetrahedron	edge a	$V=a^3/(6\sqrt{2})$; $S=\sqrt{3}\cdot a^2$; height $=a\sqrt{(2/3)}$.
Torus	major radius R from center to tube center; tube radius $r<R$	$V=2\pi^2Rr^2$; $S=4\pi^2Rr$.

Coordinate and contest geometry

Shape / rule	Meaning	Formulas
Two points	$P(x_1,y_1)$, $Q(x_2,y_2)$	distance=$\sqrt{((x_2-x_1)^2+(y_2-y_1)^2)}$; midpoint=$((x_1+x_2)/2,(y_1+y_2)/2)$.
Dot and cross	vectors u,v; angle θ	$u\cdot v=\|u\|\|v\|\cos\theta$; cross$(u,v)=u_xv_y-u_yv_x=\|u\|\|v\|\sin\theta$; sign of cross gives orientation.
Point to line	point P; line through A,B	distance=$\|\text{cross}(B-A,P-A)\|/\|B-A\|$; projection=$A+(B-A)\cdot\text{dot}(P-A,B-A)/\|B-A\|^2$.
Polygon	vertices in boundary order	signed $2A=\Sigma\ \text{cross}(P_i,P_{i+1})$; perimeter=$\Sigma\|P_{i+1}-P_i\|$; centroid=$\Sigma(P_i+P_{i+1})\text{cross}(P_i,P_{i+1})/(3\cdot\text{signed } 2A)$.
Lattice polygon	integer vertices; B boundary lattice points	Pick: $A=I+B/2-1$; $B=\Sigma\ \gcd(\|\Delta x\|,\|\Delta y\|)$ over edges.
Triangle laws	opposite sides a,b,c and angles A,B,C	$a^2=b^2+c^2-2bc\cos A$; $a/\sin A=b/\sin B=c/\sin C=2R$.
Rotation	point (x,y), angle θ counterclockwise	$(x',y')=(x\cos\theta-y\sin\theta,\ x\sin\theta+y\cos\theta)$.
Circle equations	center (h,k), radius r	$(x-h)^2+(y-k)^2=r^2$; tangent at point T is perpendicular to radius OT.

17. MATH AND CONTEST FORMULAS

Number theory and counting

Topic	Conditions	Formula
Prime factors	$n=\prod p_i^{e_i}$	divisor count $\tau(n)=\prod(e_i+1)$; divisor sum $\sigma(n)=\prod(p_i^{(e_i+1)}-1)/(p_i-1)$; $\varphi(n)=n\prod(1-1/p_i)$.
Factorials	prime p	Exponent of p in n!: $\Sigma_{k\ge1}\lfloor n/p^k\rfloor$. Trailing zeros in base b: min over $p^e\|b$ of $\lfloor v_p(n!)/e\rfloor$.
Binomial	$0\le k\le n$	$C(n,k)=n!/(k!(n-k)!)=C(n-1,k-1)+C(n-1,k)$; if p is prime, factorial/inverse method needs $n<p$.
Stars and bars	n identical items; k groups	Allow empty groups: $C(n+k-1,k-1)$. Require positive groups: $C(n-1,k-1)$, $n\ge k$.
Inclusion–exclusion	finite sets $A_1\ldots A_n$	$\|\cup A_i\|=\Sigma\|A_i\|-\Sigma\|A_i\cap A_j\|+\Sigma\|A_i\cap A_j\cap A_k\|-\cdots$.
Catalan	$n\ge0$	$\text{Cat}(n)=C(2n,n)/(n+1)$; $\text{Cat}(0)=1$; $\text{Cat}(n+1)=\Sigma_{i=0}^{n}\text{Cat}(i)\text{Cat}(n-i)$.
Repeated permutations	n items, counts $c_1\ldots c_k$	Number of distinct permutations=$n!/(c_1!\cdots c_k!)$. Derangements: $D_n=(n-1)(D_{n-1}+D_{n-2})$.
Burnside	finite symmetry group G acting on X	Orbits=$(1/\|G\|)\Sigma_{g\in G}\|\text{Fix}(g)\|$. Rotational necklaces: $(1/n)\Sigma_{d\|n}\varphi(d)\cdot\text{colors}^{(n/d)}$.
Lucas	prime modulus p; base-p digits n_i,k_i	$C(n,k)\bmod p=\prod_i C(n_i,k_i)\bmod p$.
GCD and CRT	positive moduli m,n	$ax+by=c$ solvable iff $\gcd(a,b)\|c$. $x\equiv a\pmod m$, $x\equiv b\pmod n$ compatible iff $\gcd(m,n)\|(b-a)$.
Signed division	b>0; mathematical floor/ceil	C++ integer / truncates toward zero. Normalize remainder to [0,b); $\text{floor}(a/b)=(a-r)/b$, $\text{ceil}(a/b)=\text{floor}(a/b)+(r\ne0)$.

Probability, graphs and games

Topic	Conditions	Formula
Probability	events A,B; P(B)>0	$P(A\|B)=P(A\cap B)/P(B)$; $P(A\cup B)=P(A)+P(B)-P(A\cap B)$; independent ⇒ $P(A\cap B)=P(A)P(B)$.
Expectation	random variables X,Y	$E[X]=\Sigma x\cdot P(X=x)$; $E[X+Y]=E[X]+E[Y]$ even without independence; $E[1_A]=P(A)$.
Variance	random variable X	$\text{Var}(X)=E[X^2]-E[X]^2$; independent X,Y ⇒ $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)$.
Binomial distribution	$X\sim\text{Bin}(n,p)$	$P(X=k)=C(n,k)p^k(1-p)^{(n-k)}$; $E[X]=np$; $\text{Var}(X)=np(1-p)$.
Euler trails	connected undirected graph after ignoring isolated vertices	Euler circuit iff every degree is even; Euler path iff exactly 0 or 2 vertices have odd degree.
Planar graphs	connected planar embedding	$V-E+F=2$. Simple planar $V\ge3$ ⇒ $E\le3V-6$; simple bipartite planar $V\ge3$ ⇒ $E\le2V-4$.
Bipartite matching	bipartite graph	König: maximum matching size = minimum vertex-cover size. DAG minimum path cover = V−maximum matching in split graph.
Grundy	finite impartial game with normal play	$g(v)=\text{mex}\{g(u): v\to u\}$; disjoint games combine by XOR; losing position iff combined Grundy=0.